

EUROPEAN
SPALLATION
SOURCE

LENS WG4 · Garching · 7 Oct 2026

Andrew Sazonov

European Spallation Source (ESS)
Data Management and Scientific Computing (DMSC)

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EasyDiffraction and crysta

Reusing crystallographic libraries, and building our own

Andrew Sazonov

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Data Management and Scientific Computing (DMSC)

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1 WHY NEW SOFTWARE

1 WHY NEW SOFTWARE

2 EASYDIFFRACTION

1 WHY NEW SOFTWARE

2 EASYDIFFRACTION

3 CRYSTA

1 WHY NEW SOFTWARE

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3 CRYSTA

4 STATUS AND PLANS

1 WHY NEW SOFTWARE
2 EASYDIFFRACTION
3 CRYSTA
4 STATUS AND PLANS

1.1 WHY NEW SOFTWARE

Diffraction analysis: many tools

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DIFFRACTION PROGRAMS

FullProf Suite · Jana2020 · GSAS-II · SHELX · MAUD · FOX · Mag2Pol · TOPAS · HighScore · PDFgui · …

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DIFFRACTION LIBRARIES

CrysFML · CCSL · cctbx · pyobjcryst · CrysPy · PDFfit2 · GSAS-II · …

1.1 WHY NEW SOFTWARE

Diffraction analysis: many tools

Many tools, each with its own interface.

DIFFRACTION PROGRAMS

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1.2 WHY NEW SOFTWARE

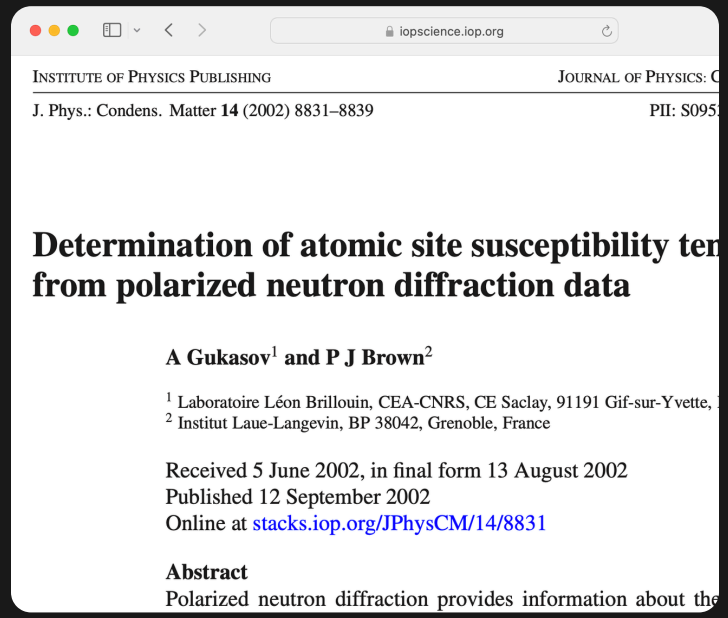
What users run into

1.2 WHY NEW SOFTWARE

What users run into

Site susceptibility

CrysPy, CCSL only · for MAGIC



iopscience.iop.org

INSTITUTE OF PHYSICS PUBLISHING — JOURNAL OF PHYSICS: C

J. Phys.: Condens. Matter **14** (2002) 8831–8839 — PII: S095

Determination of atomic site susceptibility ten from polarized neutron diffraction data

A Gukasov[1] **and P J Brown**[2]

[1] Laboratoire Léon Brillouin, CEA-CNRS, CE Saclay, 91191 Gif-sur-Yvette,
[2] Institut Laue-Langevin, BP 38042, Grenoble, France

Received 5 June 2002, in final form 13 August 2002
Published 12 September 2002
Online at stacks.iop.org/JPhysCM/14/8831

Abstract

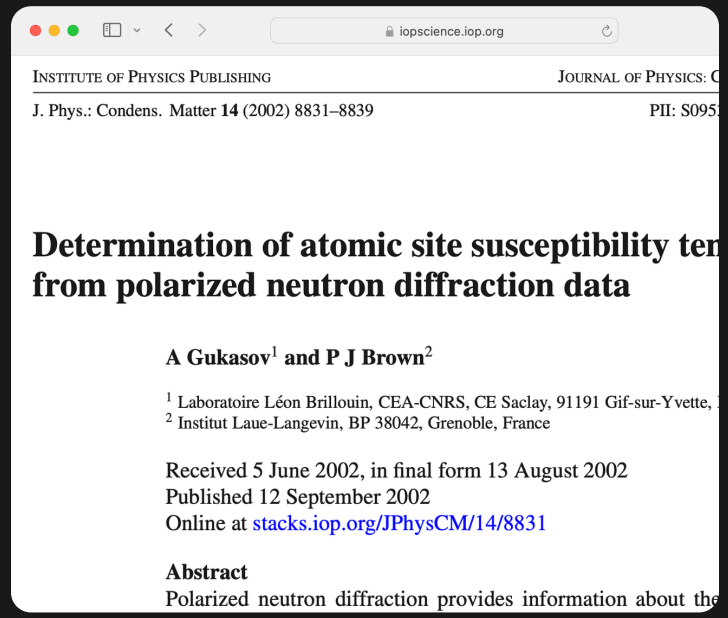
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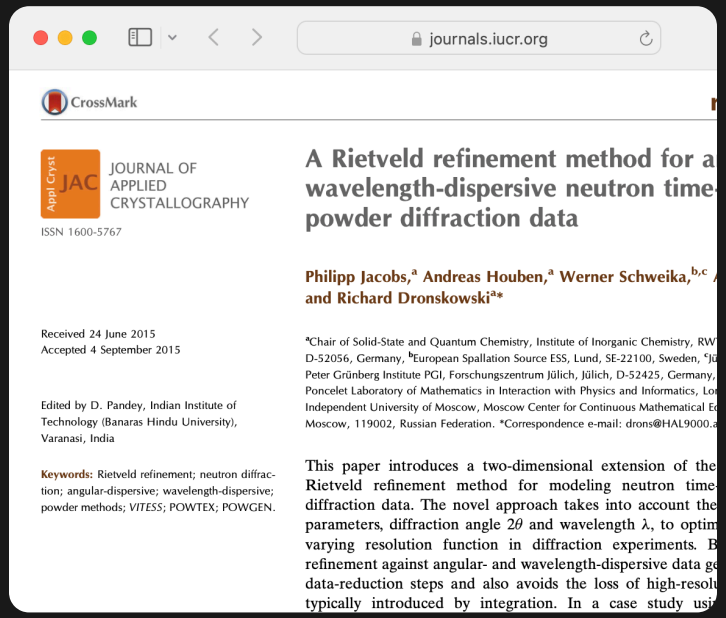
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Abstract
Polarized neutron diffraction provides information about the

2D Rietveld

modified GSAS-II only · for DREAM



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JOURNAL OF APPLIED CRYSTALLOGRAPHY

ISSN 1600-5767

A Rietveld refinement method for a wavelength-dispersive neutron time- powder diffraction data

Philipp Jacobs,[a] Andreas Houben,[a] Werner Schweika,[b,c] and Richard Dronskowski[a]*

[a]Chair of Solid-State and Quantum Chemistry, Institute of Inorganic Chemistry, RW D-52056, Germany, [b]European Spallation Source ESS, Lund, SE-22100, Sweden, [c]Jü Peter Grünberg Institute PGI, Forschungszentrum Jülich, Jülich, D-52425, Germany, Poncelet Laboratory of Mathematics in Interaction with Physics and Informatics, Lo Independent University of Moscow, Moscow Center for Continuous Mathematical Ed Moscow, 119002, Russian Federation. *Correspondence e-mail: drons@HAL9000.a

Received 24 June 2015
Accepted 4 September 2015

Edited by D. Pandey, Indian Institute of Technology (Banaras Hindu University), Varanasi, India

Keywords: Rietveld refinement; neutron diffraction; angular-dispersive; wavelength-dispersive; powder methods; *VITESS*; POWTEX; POWGEN.

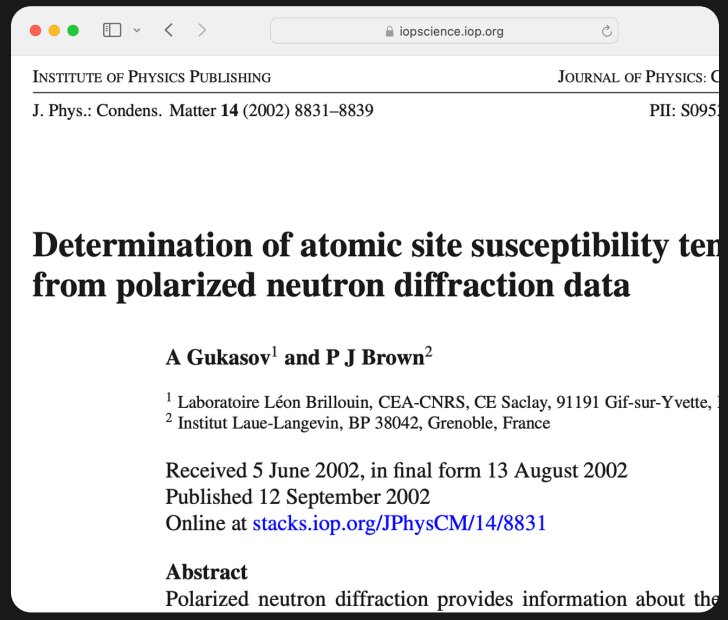
This paper introduces a two-dimensional extension of the Rietveld refinement method for modeling neutron time- diffraction data. The novel approach takes into account the parameters, diffraction angle 2θ and wavelength λ, to optim varying resolution function in diffraction experiments. B refinement against angular- and wavelength-dispersive data ge data-reduction steps and also avoids the loss of high-resolu typically introduced by integration. In a case study usi

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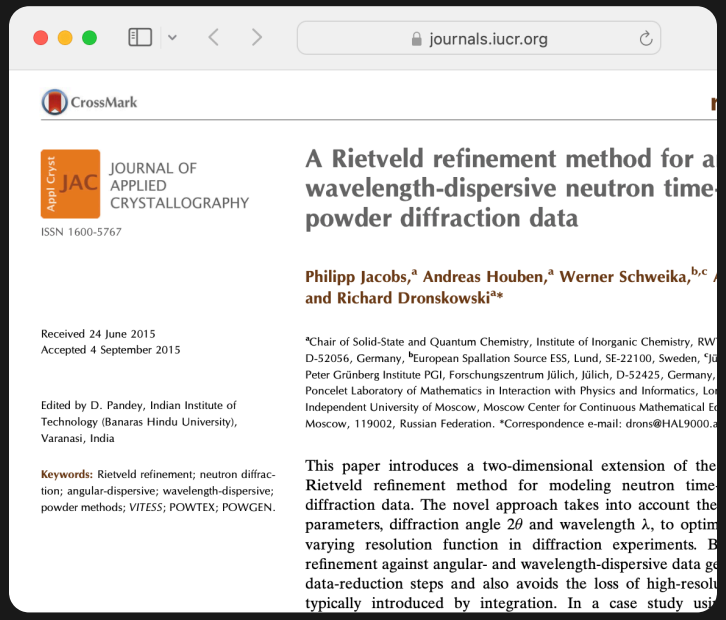
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Appl Cryst JAC JOURNAL OF APPLIED CRYSTALLOGRAPHY

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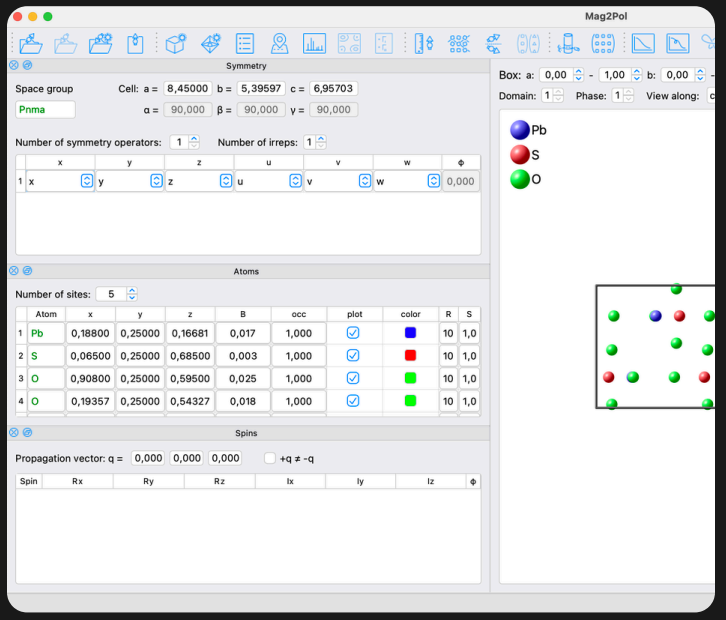
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Not the best UX

many panels and controls at once

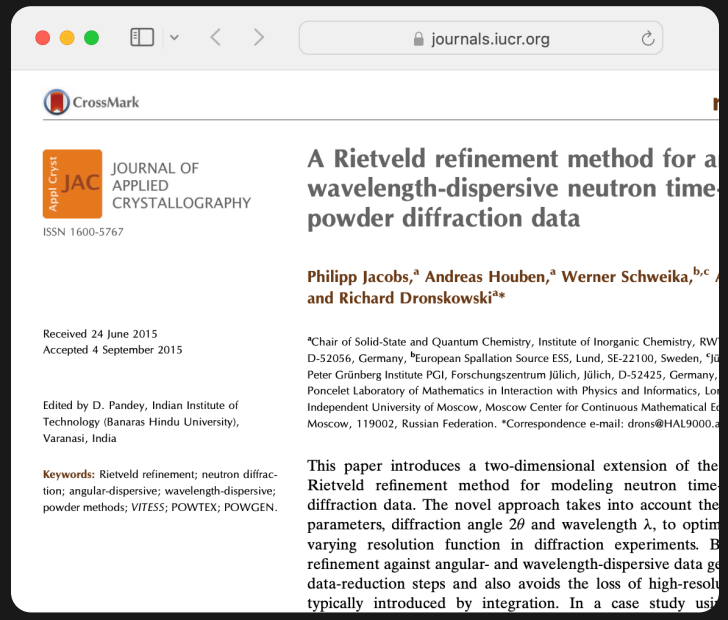


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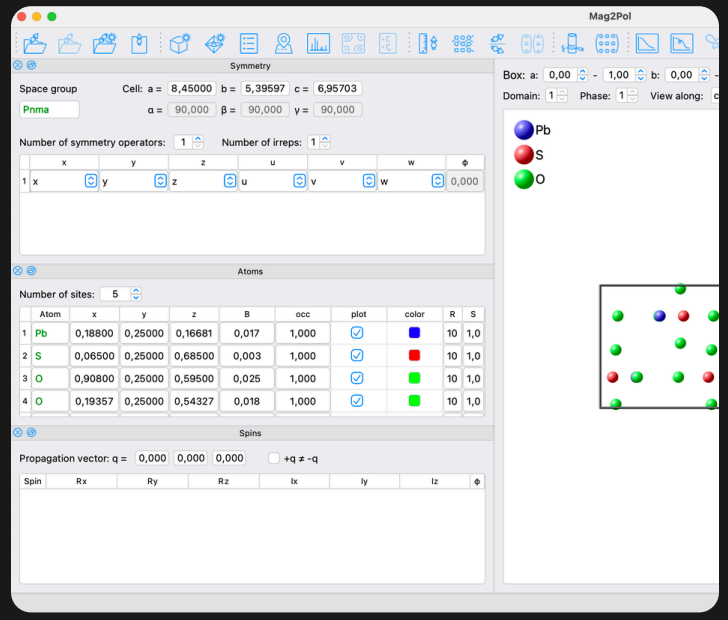
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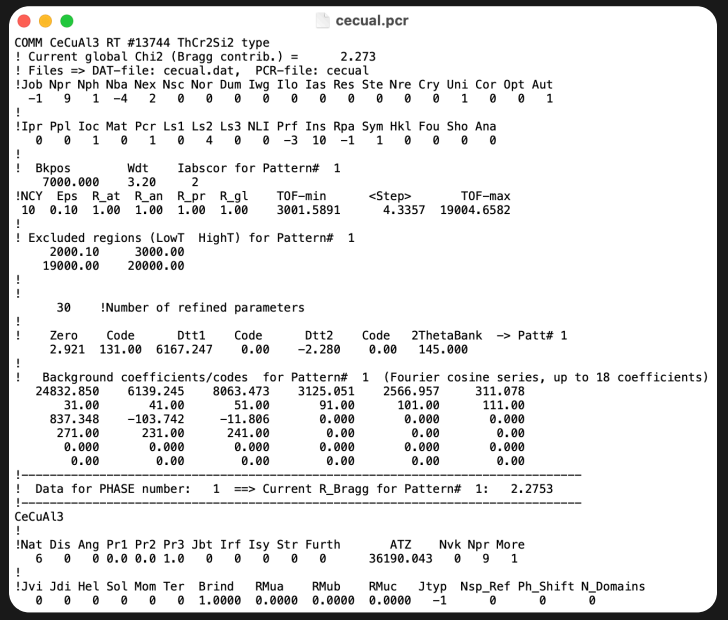
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many panels and controls at once



High entrance barrier

input files with cryptic labels

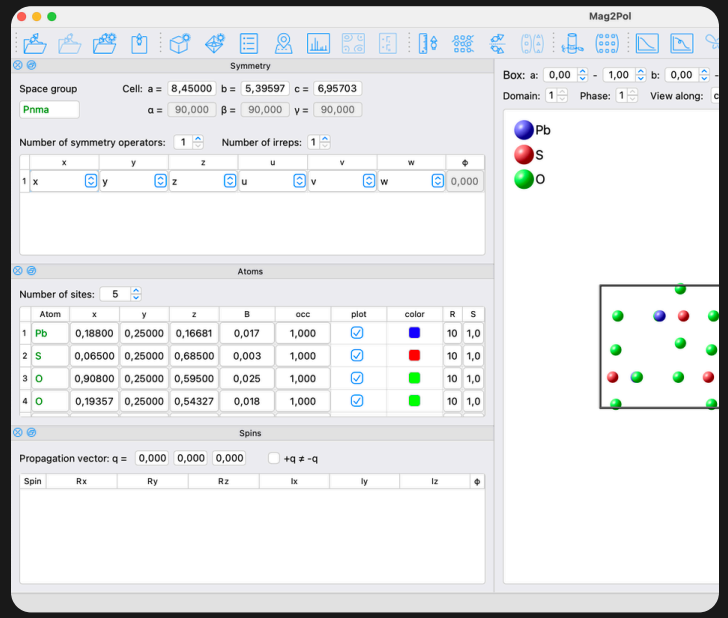


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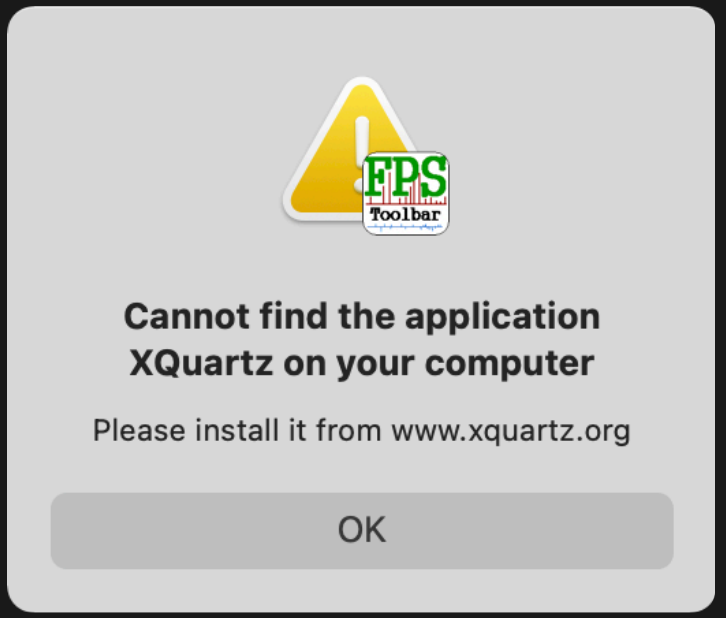
input files with cryptic labels

cecual.pcr

```
COMM CeCuAl3 RT #13744 ThCr2Si2 type
! Current global Chi2 (Bragg contrib.) =      2.273
! Files => DAT-file: cecual.dat,  PCR-file: cecual
!Job Npr Nph Nba Nex Nsc Nor Dum Iwg Ilo Ias Res Ste Nre Cry Uni Cor Opt Aut
  -1   9   1  -4   2   0   0   0   0   0   0   0   0   0   1   0   0   1
!
!Ipr Ppl Ioc Mat Pcr Ls1 Ls2 Ls3 NLI Prf Ins Rpa Sym Hkl Fou Sho Ana
   0   0   1   0   1   0   4   0   0  -3  10  -1   1   0   0   0   0
!
!  Bkpos        Wdt     Iabscor for Pattern#  1
    7000.000    3.20       2
!NCY  Eps  R_at  R_an  R_pr  R_gl    TOF-min      <Step>       TOF-max
 10  0.10  1.00  1.00  1.00  1.00    3001.5891      4.3357  19004.6582
!
! Excluded regions (LowT  HighT) for Pattern#  1
     2000.10     3000.00
    19000.00    20000.00
!
!
      30    !Number of refined parameters
!
!    Zero     Code      Dtt1     Code      Dtt2     Code   2ThetaBank  -> Patt# 1
     2.921  131.00  6167.247     0.00    -2.280     0.00   145.000
!
!   Background coefficients/codes  for Pattern#  1  (Fourier cosine series, up to 18 coefficients)
   24832.850    6139.245    8063.473    3125.051    2566.957     311.078
      31.00       41.00       51.00       91.00      101.00      111.00
     837.348    -103.742    -11.806       0.000       0.000       0.000
     271.00      231.00      241.00        0.00        0.00        0.00
       0.000       0.000       0.000       0.000       0.000       0.000
       0.00        0.00        0.00        0.00        0.00        0.00
!-------------------------------------------------------------------------------
!  Data for PHASE number:   1  ==> Current R_Bragg for Pattern#  1:    2.2753
!-------------------------------------------------------------------------------
CeCuAl3
!
!Nat Dis Ang Pr1 Pr2 Pr3 Jbt Irf Isy Str Furth       ATZ     Nvk Npr More
   6   0   0 0.0 0.0 1.0   0   0   0   0   0      36190.043   0   9   1
!
!Jvi Jdi Hel Sol Mom Ter  Brind   RMua    RMub    RMuc   Jtyp  Nsp_Ref Ph_Shift N_Domains
   0   0   0   0   0   0 1.0000  0.0000  0.0000  0.0000   -1      0       0        0
```

Extra dependencies

XQuartz to install on macOS



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!Job Npr Nph Nba Nex Nsc Nor Dum Iwg Ilo Ias Res Ste Nre Cry Uni Cor Opt Aut
  -1   9   1  -4   2   0   0   0   0   0   0   0   0   0   1   0   0   1
!
!Ipr Ppl Ioc Mat Pcr Ls1 Ls2 Ls3 NLI Prf Ins Rpa Sym Hkl Fou Sho Ana
   0   0   1   0   1   0   4   0   0  -3  10  -1   1   0   0   0   0
!
!  Bkpos        Wdt    Iabscor for Pattern#  1
    7000.000    3.20     2
!NCY  Eps  R_at  R_an  R_pr  R_gl    TOF-min      <Step>       TOF-max
 10  0.10  1.00  1.00  1.00  1.00    3001.5891     4.3357  19004.6582
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! Excluded regions (LowT  HighT) for Pattern#  1
    2000.10    3000.00
   19000.00   20000.00
!
!
      30    !Number of refined parameters
!
!    Zero     Code      Dtt1     Code      Dtt2     Code   2ThetaBank  -> Patt# 1
    2.921  131.00  6167.247    0.00    -2.280    0.00   145.000
!
!   Background coefficients/codes  for Pattern#  1  (Fourier cosine series, up to 18 coefficients)
  24832.850    6139.245    8063.473    3125.051    2566.957     311.078
     31.00       41.00       51.00       91.00      101.00      111.00
    837.348    -103.742     -11.806       0.000       0.000       0.000
    271.00      231.00      241.00        0.00        0.00        0.00
      0.000       0.000       0.000       0.000       0.000       0.000
      0.00        0.00        0.00        0.00        0.00        0.00
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CeCuAl3
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!Nat Dis Ang Pr1 Pr2 Pr3 Jbt Irf Isy Str Furth       ATZ    Nvk Npr More
   6   0   0 0.0 0.0 1.0   0   0   0   0   0     36190.043   0   9   1
!
!Jvi Jdi Hel Sol Mom Ter  Brind   RMua    RMub    RMuc   Jtyp  Nsp_Ref Ph_Shift N_Domains
   0   0   0   0   0   0  1.0000  0.0000  0.0000  0.0000   -1       0       0       0
```

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Not cross-platform

no macOS or Linux support

	Download from
Jana2020 for Windows	jana2020Inst.msi
Jana2006 for Windows, STABLE	janainst.msi
Jana2006 for Unix, STABLE	not available
Jana2006 for Windows, LATEST	janainst.msi
Jana2006 for Unix, LATEST	not available

What users run into

Gaps for the ESS instruments, and a steep start for new users.

High entrance barrier

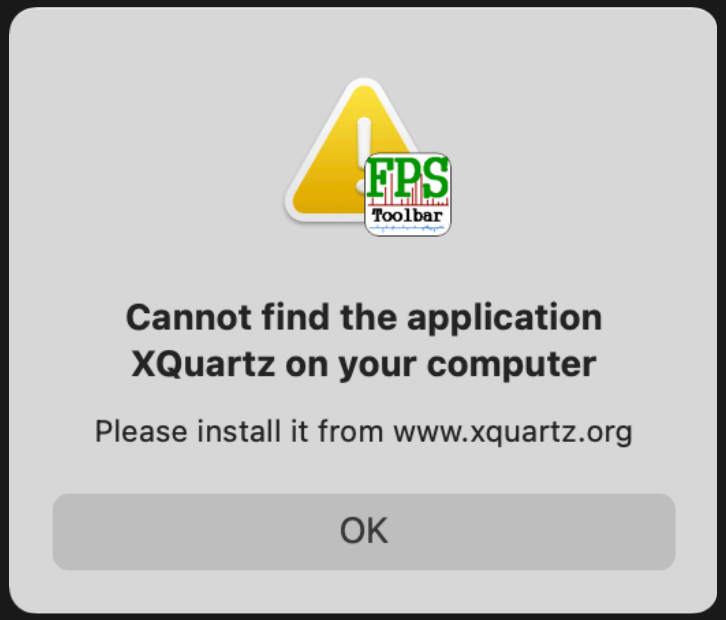
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! Files => DAT-file: cecual.dat,  PCR-file: cecual
!Job Npr Nph Nba Nex Nsc Nor Dum Iwg Ilo Ias Res Ste Nre Cry Uni Cor Opt Aut
  -1   9   1  -4   2   0   0   0   0   0   0   0   0   0   1   0   0   1
!
!Ipr Ppl Ioc Mat Pcr Ls1 Ls2 Ls3 NLI Prf Ins Rpa Sym Hkl Fou Sho Ana
   0   0   1   0   1   0   4   0   0  -3  10  -1   1   0   0   0   0
!
!  Bkpos        Wdt    Iabscor for Pattern#  1
    7000.000    3.20     2
!NCY  Eps  R_at  R_an  R_pr  R_gl    TOF-min      <Step>       TOF-max
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! Excluded regions (LowT  HighT) for Pattern#  1
    2000.10     3000.00
   19000.00    20000.00
!
!
     30    !Number of refined parameters
!
!    Zero    Code      Dtt1    Code      Dtt2    Code   2ThetaBank  -> Patt# 1
    2.921  131.00  6167.247    0.00    -2.280    0.00   145.000
!
!   Background coefficients/codes  for Pattern#  1  (Fourier cosine series, up to 18 coefficients)
   24832.850    6139.245    8063.473    3125.051    2566.957     311.078
      31.00        41.00       51.00       91.00       101.00       111.00
     837.348    -103.742    -11.806       0.000       0.000        0.000
     271.00       231.00      241.00        0.00        0.00         0.00
      0.000        0.000       0.000       0.000       0.000        0.000
       0.00         0.00        0.00        0.00        0.00         0.00
!-------------------------------------------------------------------------------
!  Data for PHASE number:   1  ==> Current R_Bragg for Pattern#  1:   2.2753
!-------------------------------------------------------------------------------
CeCuAl3
!
!Nat Dis Ang Pr1 Pr2 Pr3 Jbt Irf Isy Str Furth       ATZ    Nvk Npr More
   6   0   0 0.0 0.0 1.0   0   0   0   0   0      36190.043   0   9   1
!
!Jvi Jdi Hel Sol Mom Ter  Brind   RMua    RMub    RMuc   Jtyp  Nsp_Ref Ph_Shift N_Domains
   0   0   0   0   0   0  1.0000  0.0000  0.0000  0.0000   -1      0       0        0
```

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Not cross-platform

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	Download from
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Jana2006 for Windows, STABLE	janainst.msi
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1.3 WHY NEW SOFTWARE

Needs

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- Extend functionality for the ESS instruments

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- Extend functionality for the ESS instruments
- Improve usability for every user

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- Extend functionality for the ESS instruments
- Improve usability for every user
- Simplify maintenance for a small team

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Needs

No existing program offers all three.

- Extend functionality for the ESS instruments
- Improve usability for every user
- Simplify maintenance for a small team

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2.1 EASYDIFFRACTION

Built on existing libraries

We start from the libraries, not the programs.

DIFFRACTION PROGRAMS

FullProf Suite · Jana2020 · GSAS-II · SHELX · MAUD · FOX · Mag2Pol · TOPAS · HighScore · PDFgui · …

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CrysFML · CCSL · cctbx · pyobjcryst · CrysPy · PDFfit2 · GSAS-II · …

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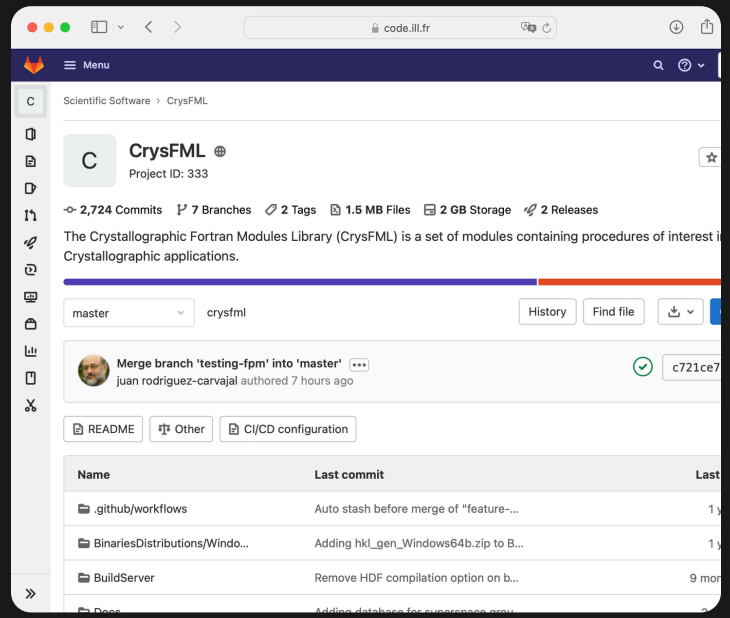
What each library brings

2.2 EASYDIFFRACTION

What each library brings

CrysFML

mature, wide coverage · Fortran

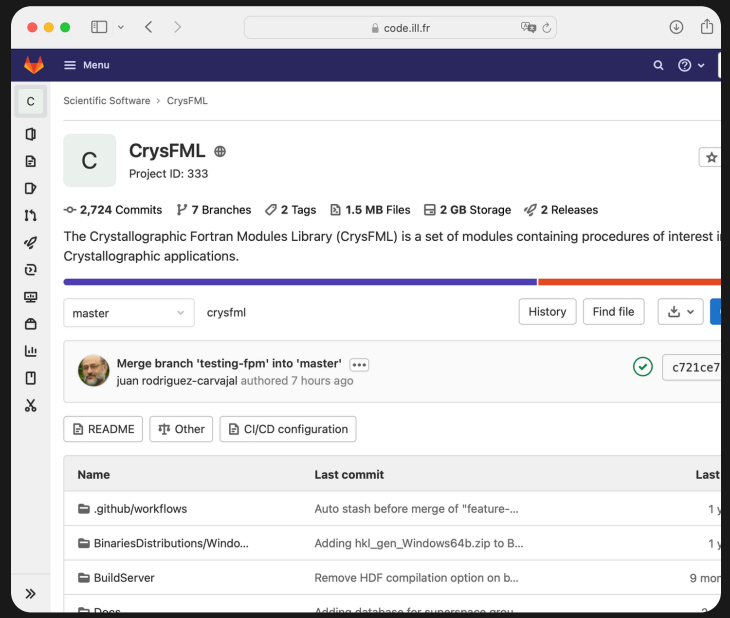


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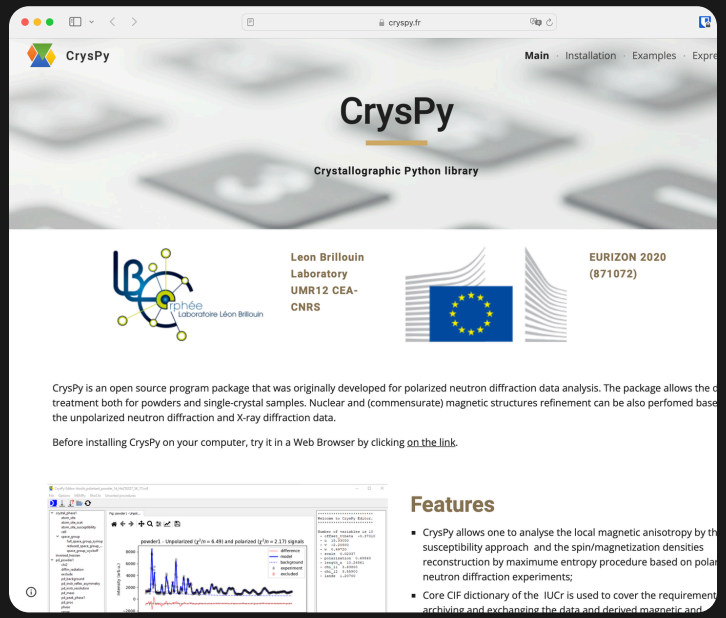
CrysFML

mature, wide coverage · Fortran



CrysPy

more polarised-neutron options · Python

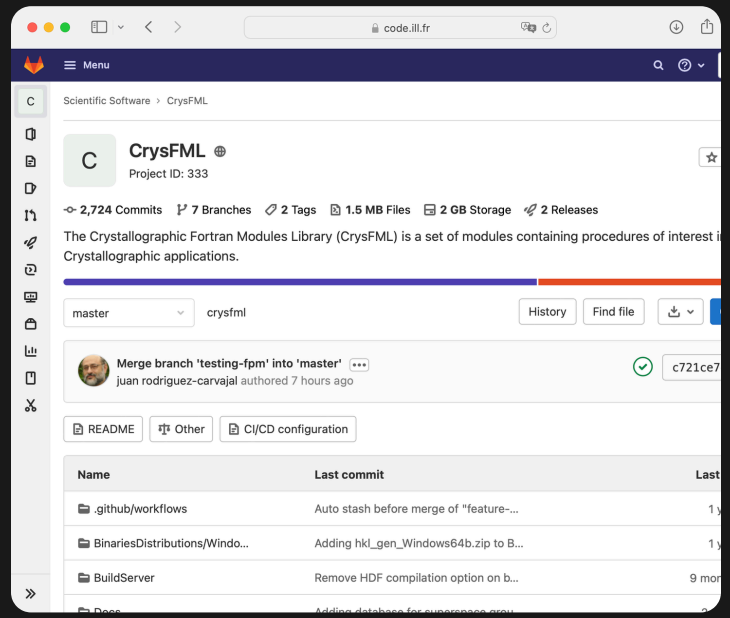


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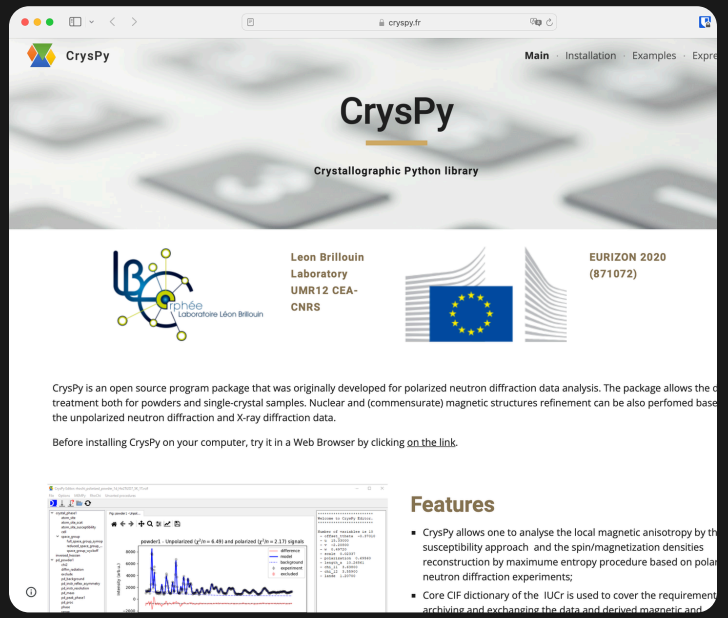
CrysFML

mature, wide coverage · Fortran



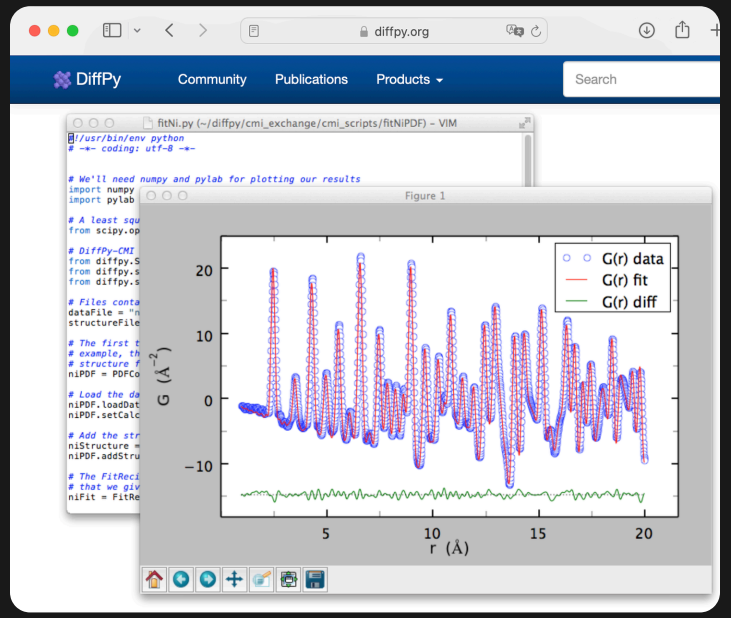
CrysPy

more polarised-neutron options · Python



PDFfit2

pair distribution function · C++

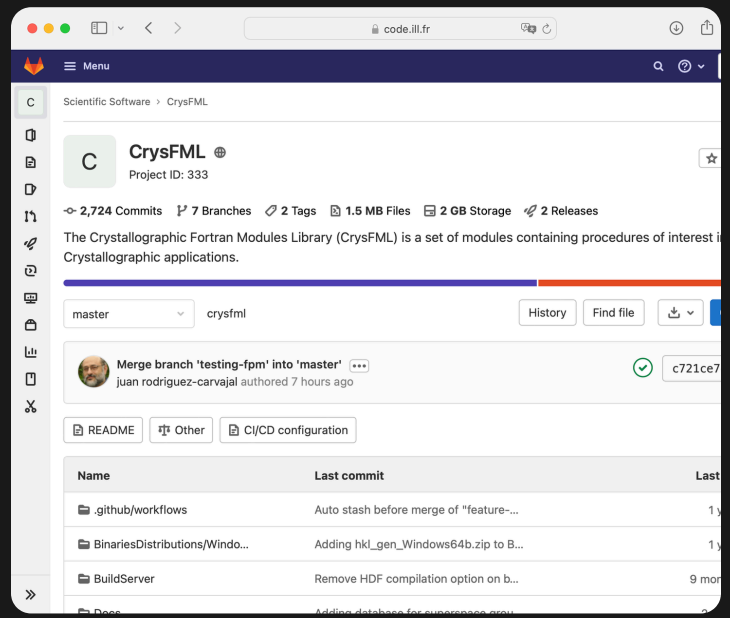


What each library brings

Each for the functionality we need.

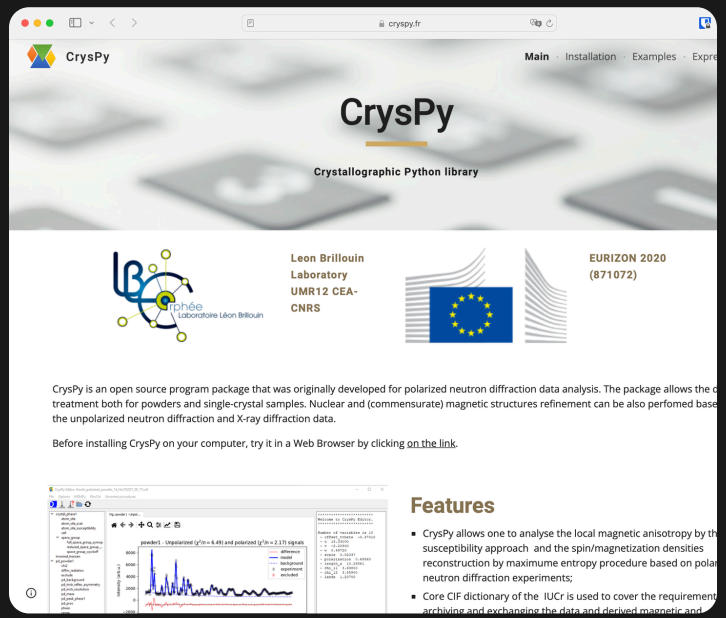
CrysFML

mature, wide coverage · Fortran



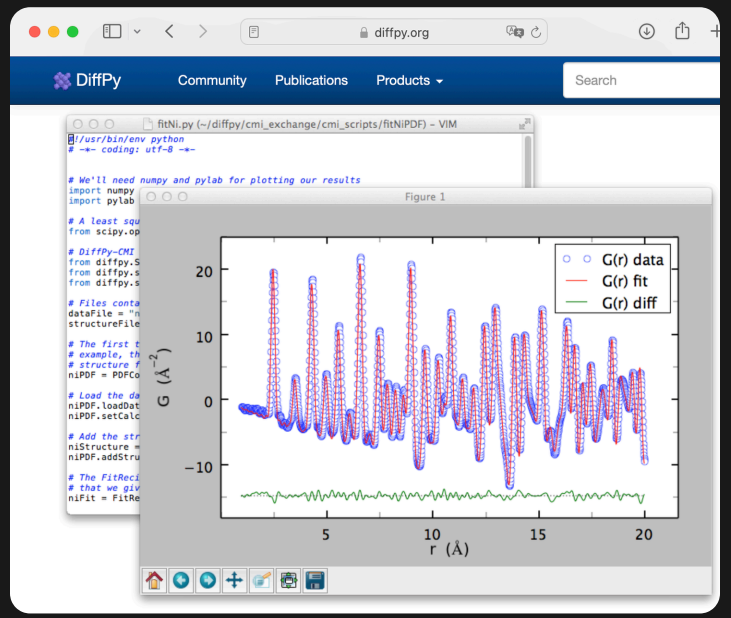
CrysPy

more polarised-neutron options · Python



PDFfit2

pair distribution function · C++



2.3 EASYDIFFRACTION

What we rely on

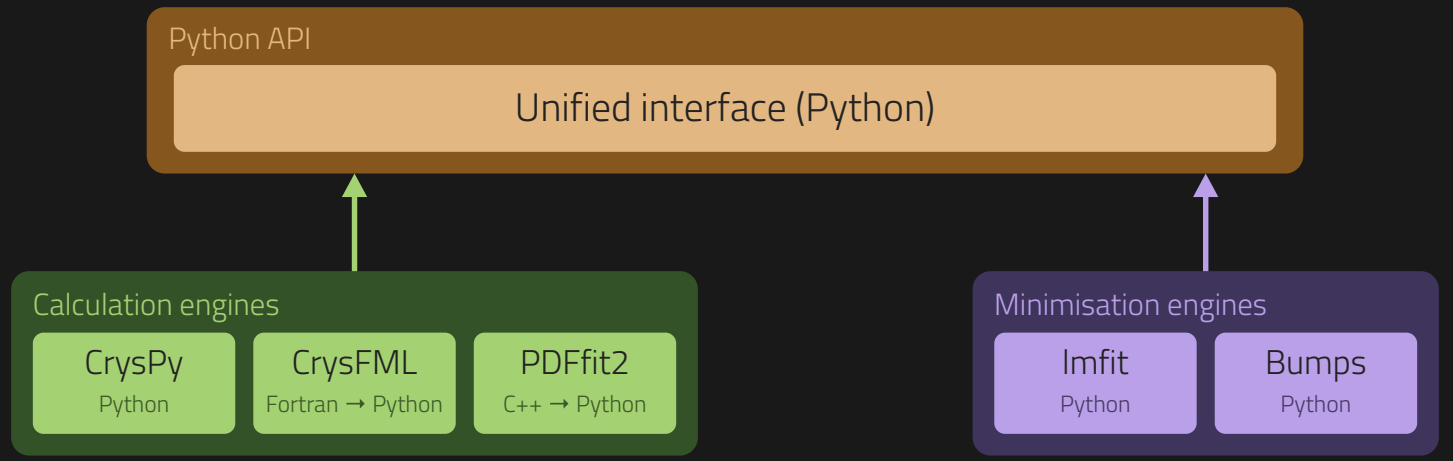
What we rely on



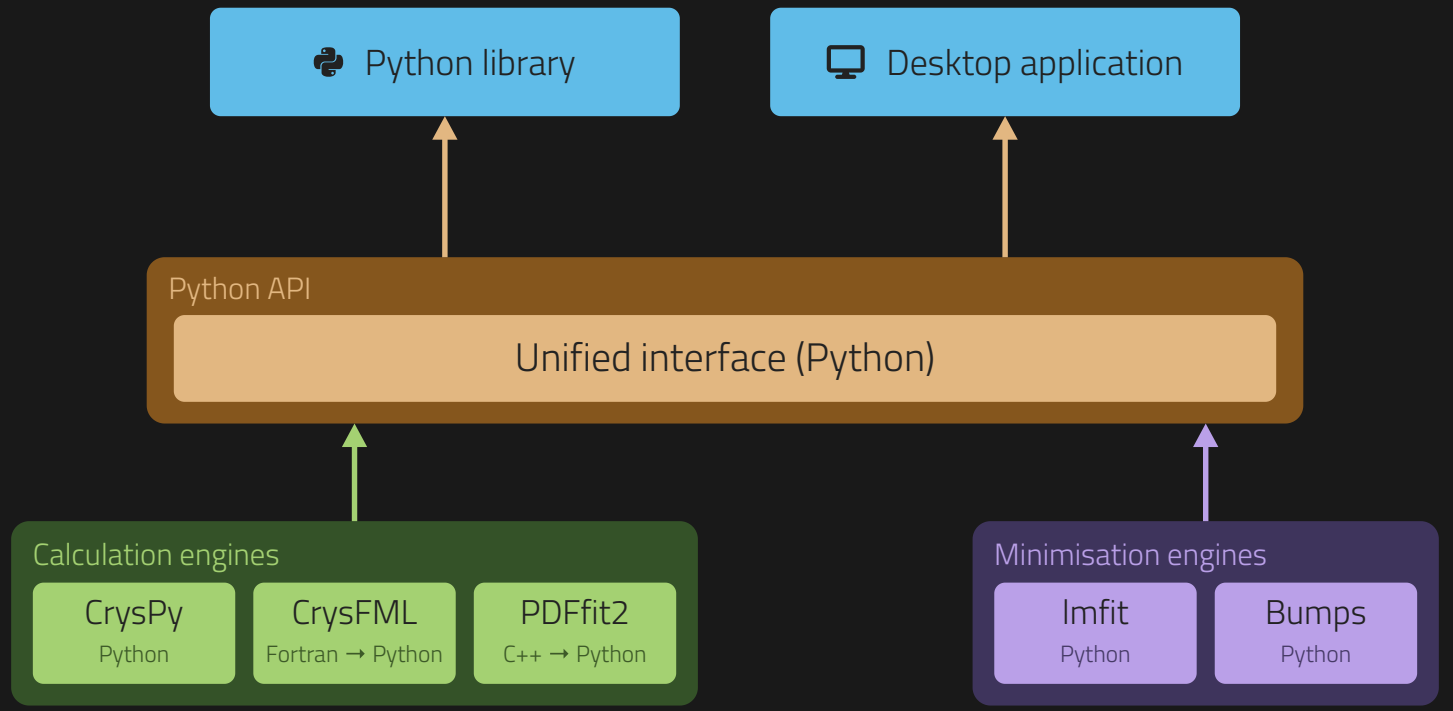
What we rely on



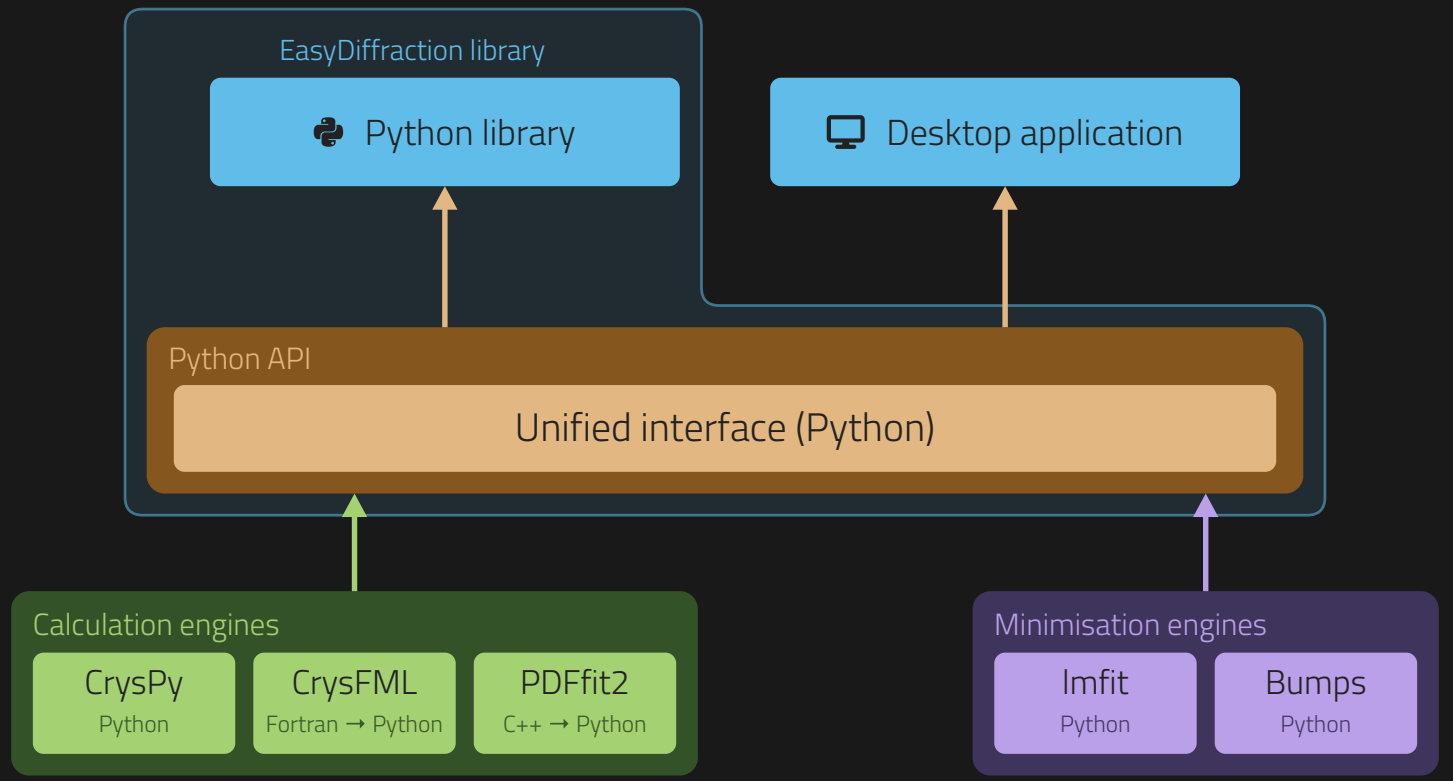
What we rely on



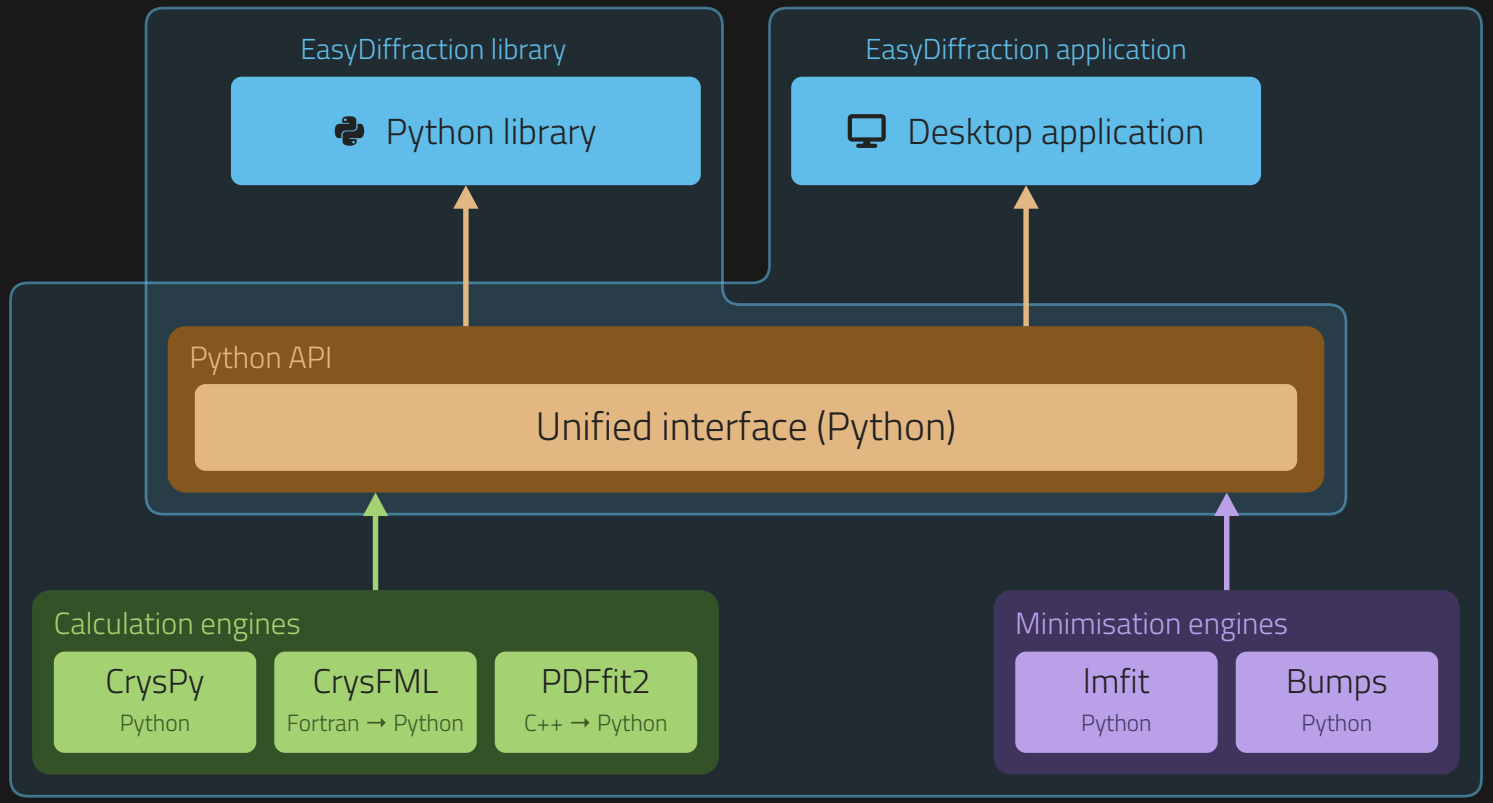
What we rely on



What we rely on

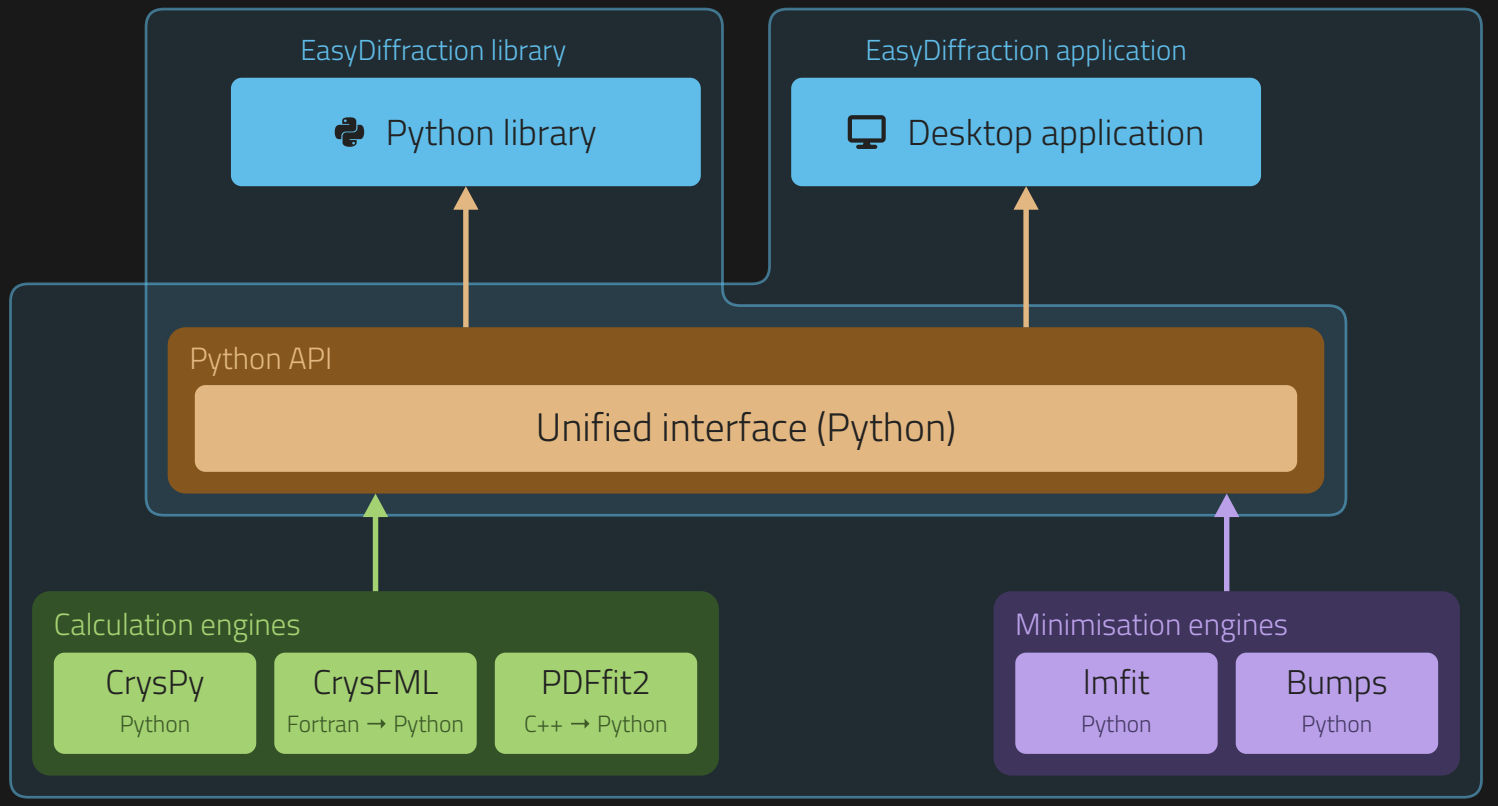


What we rely on



What we rely on

Calculations come from the libraries; the interface is ours.

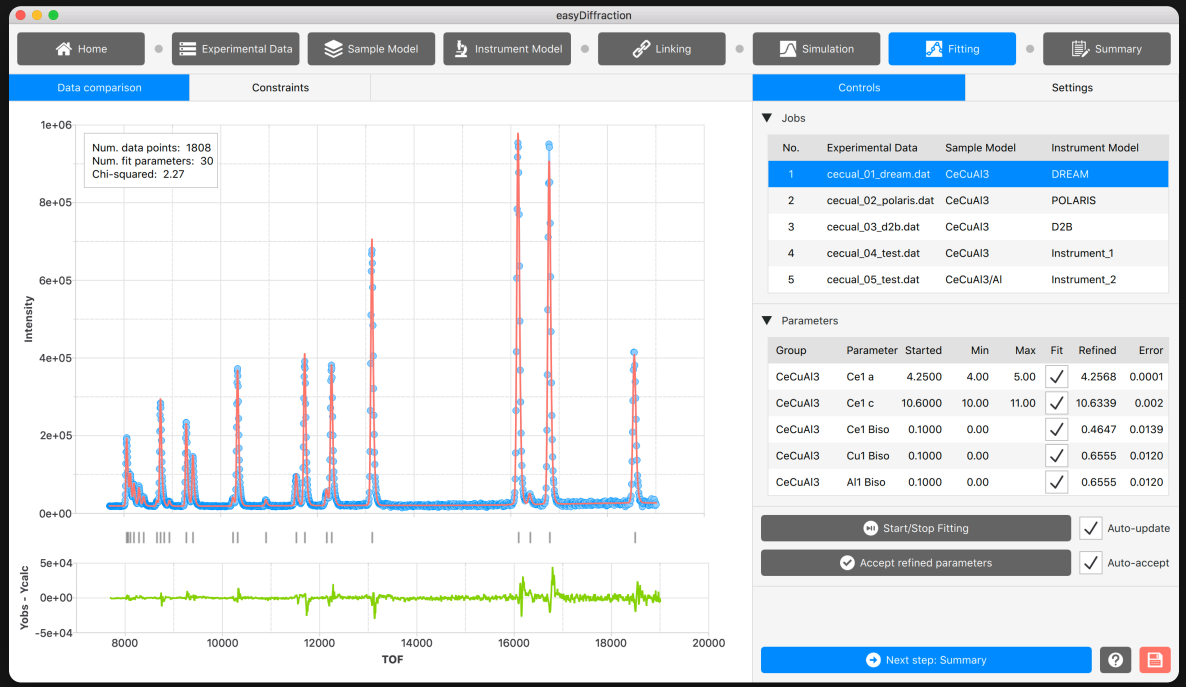


2.4 EASYDIFFRACTION

Several years of improving UI/UX

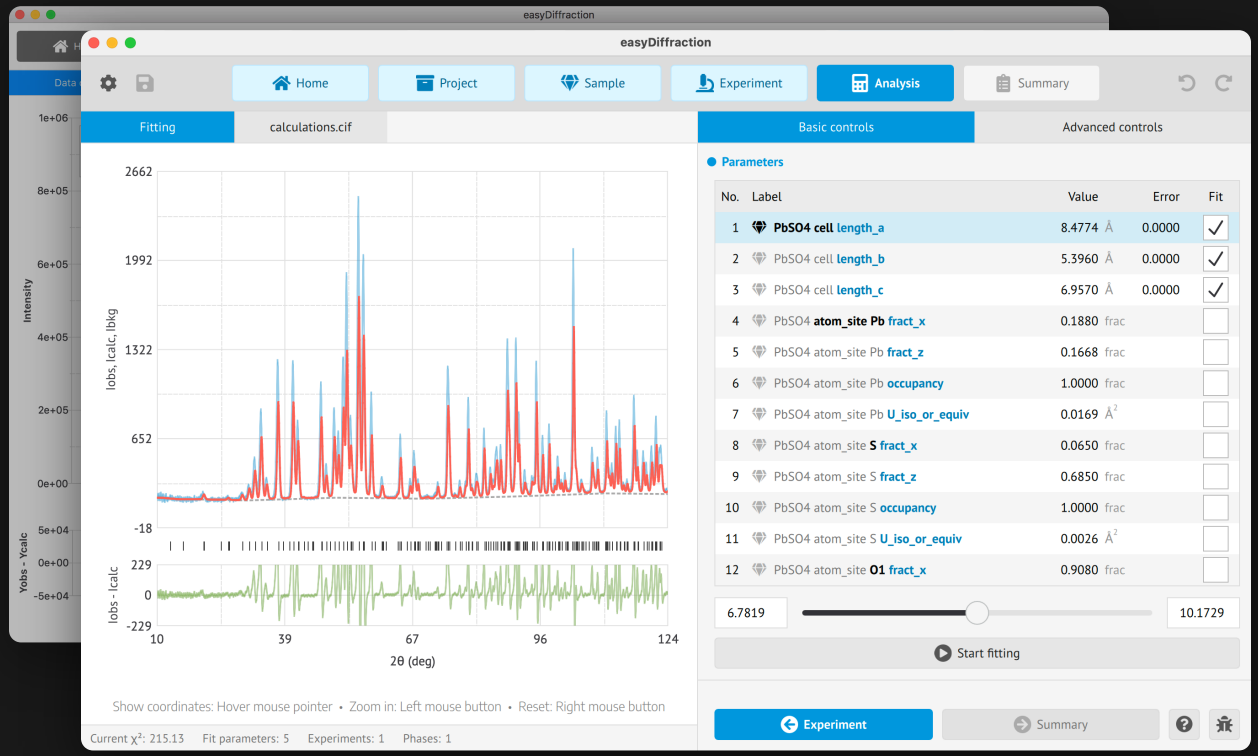
Several years of improving UI/UX

2019 · the first mock-up, for diffraction



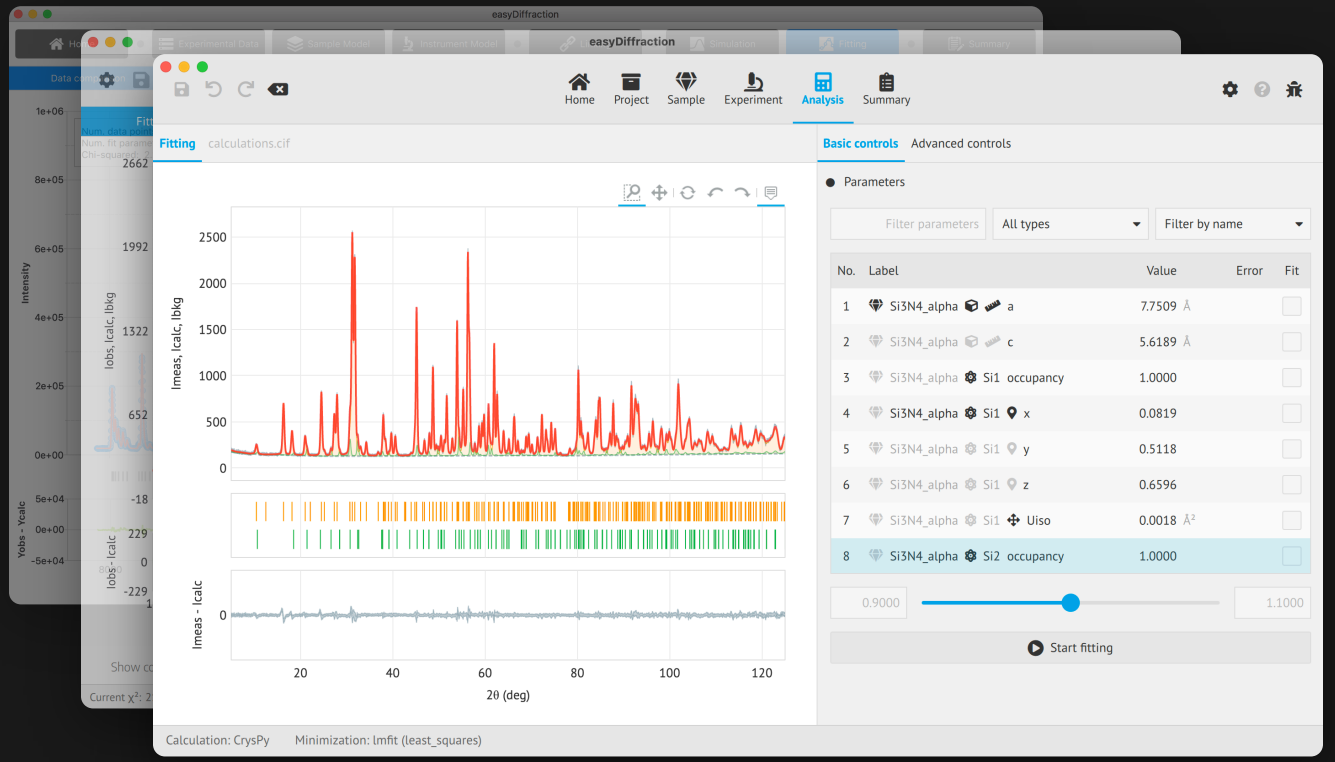
Several years of improving UI/UX

2020 · a simpler workflow



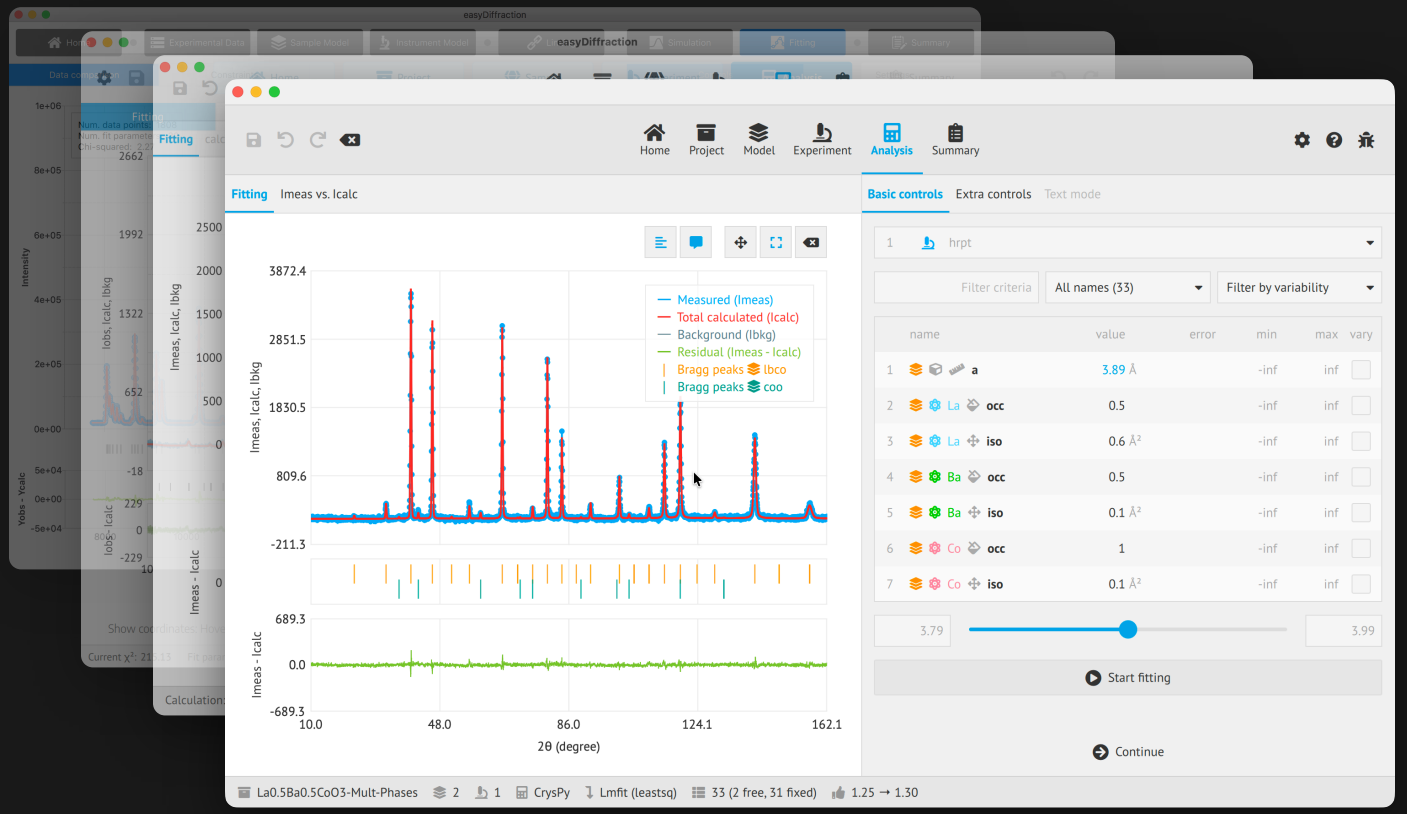
Several years of improving UI/UX

2021 · cleaner, lighter



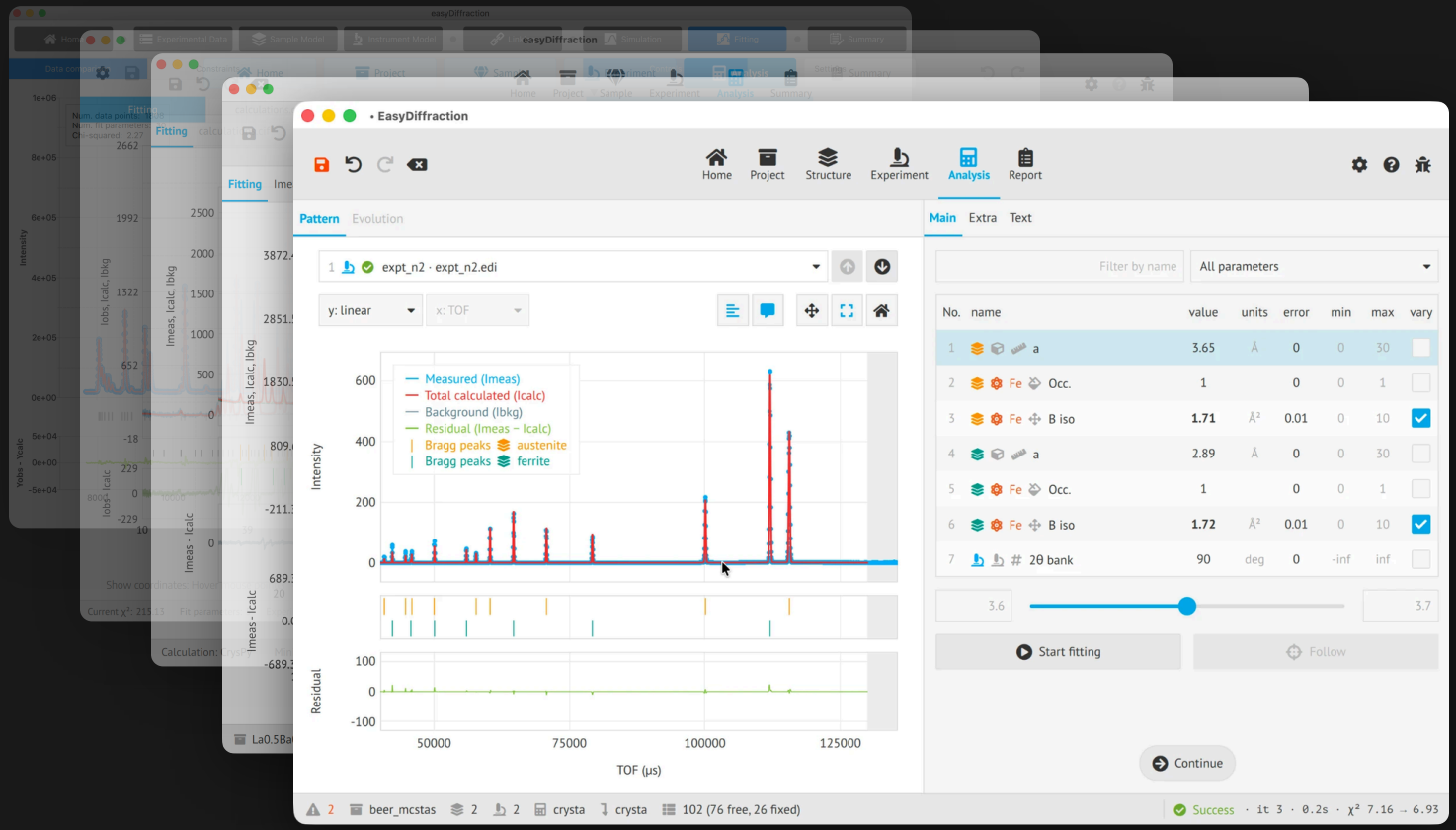
Several years of improving UI/UX

2023–2025 · the same shape, easier to use



Several years of improving UI/UX

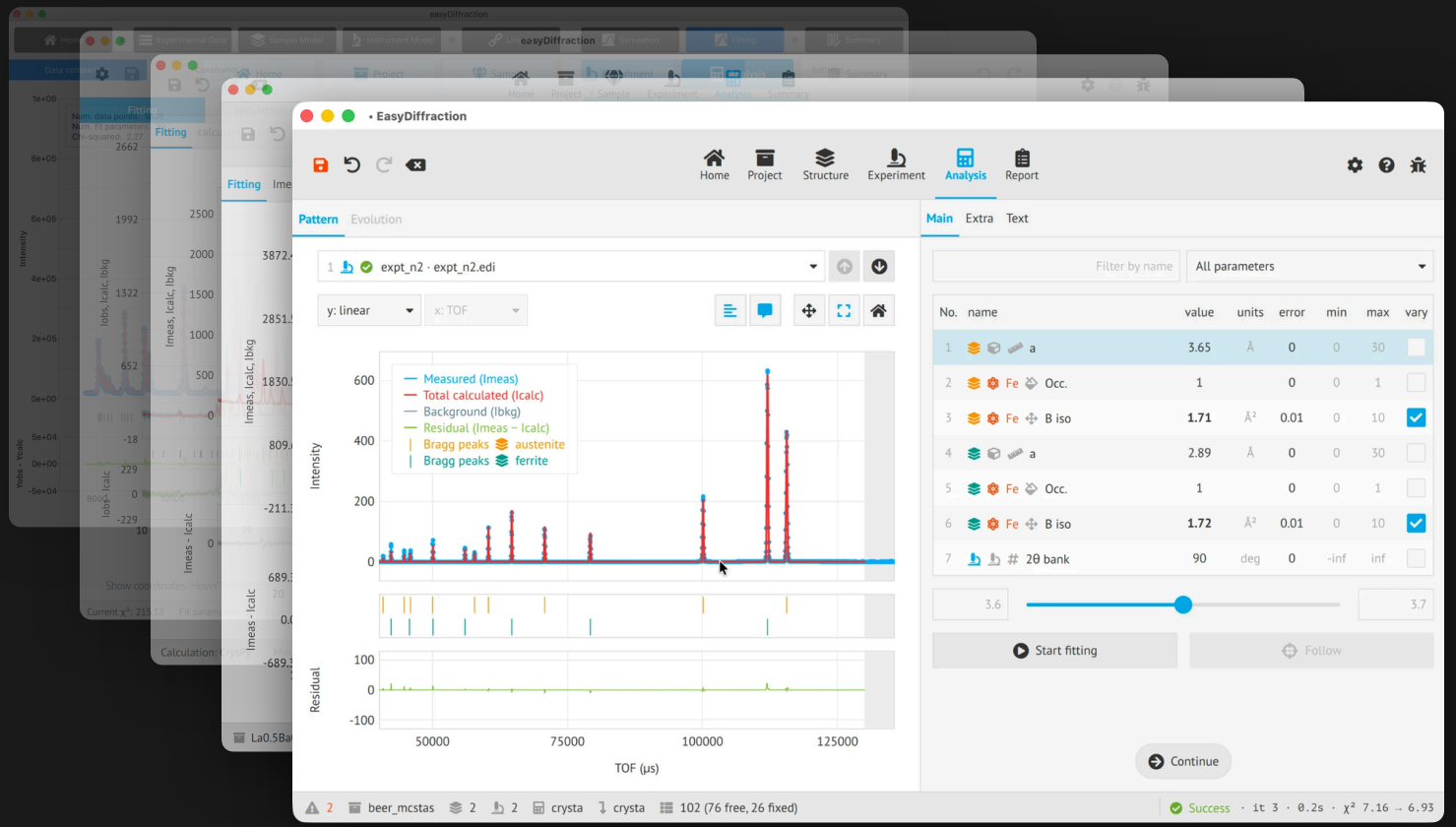
2026 · more functionality, still intuitive



Several years of improving UI/UX

The functionality grew; the user interface stayed simple.

2026 · more functionality, still intuitive

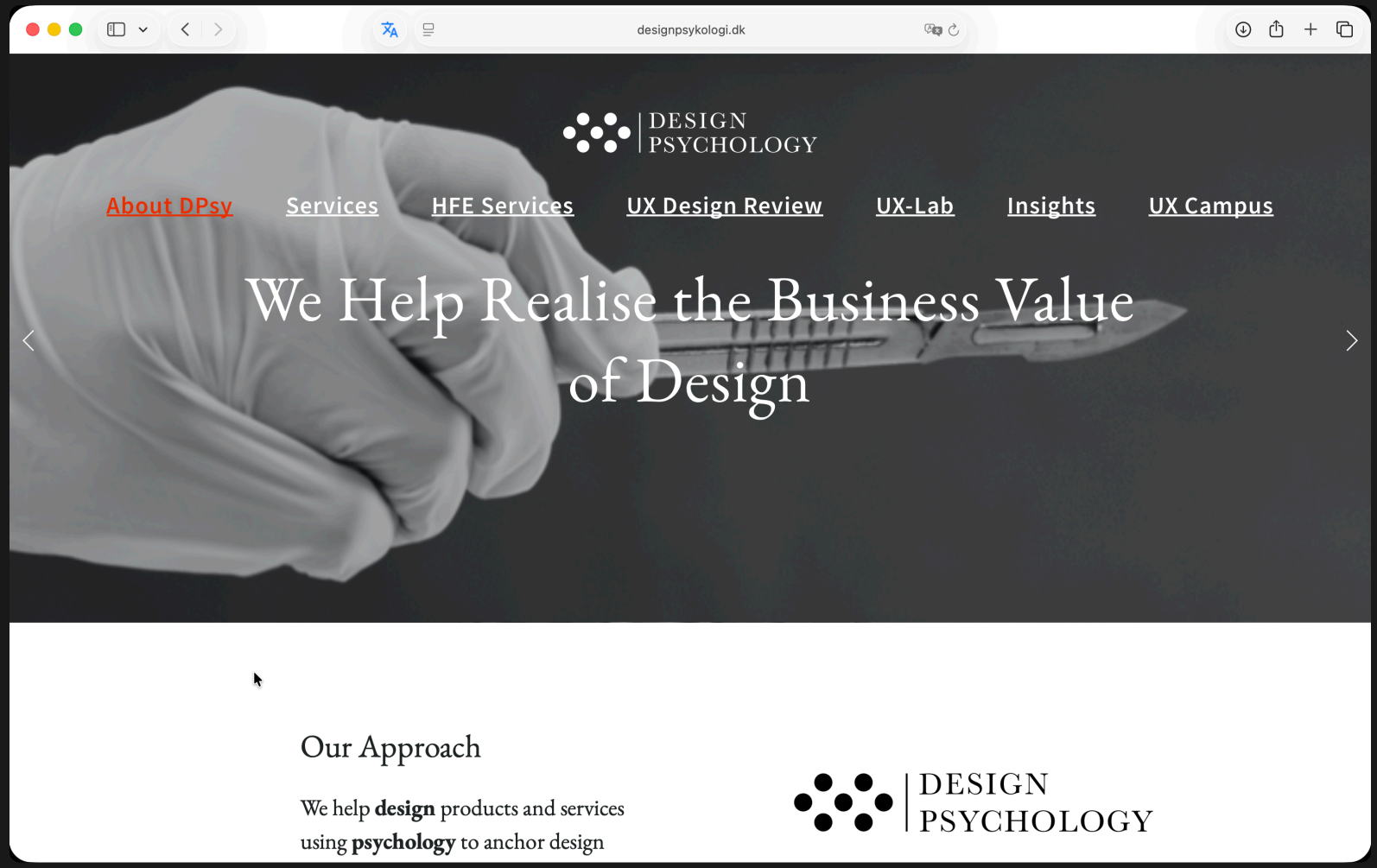


2.5 EASYDIFFRACTION

Reviewed by UX experts

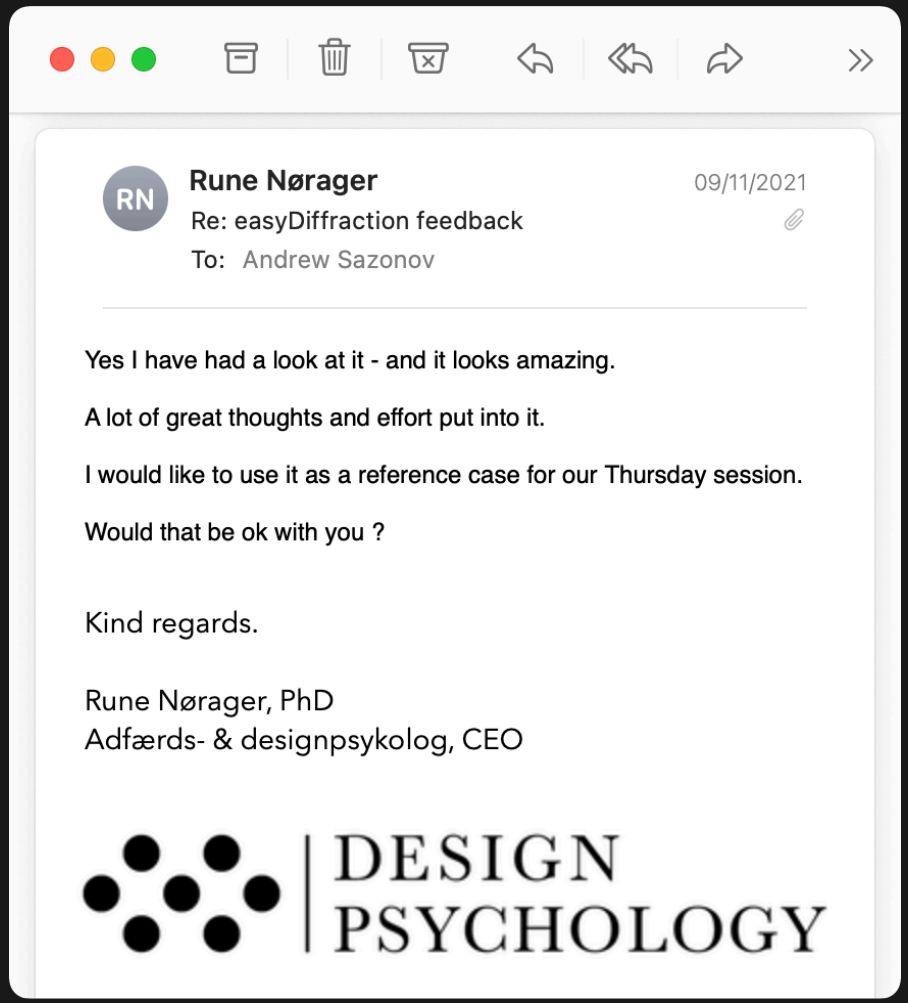
Reviewed by UX experts

Two workshops with Design Psychology, a UX consultancy.



Reviewed by UX experts

Then they reviewed our interface.



Rune Nørager 09/11/2021
Re: easyDiffraction feedback
To: Andrew Sazonov

Yes I have had a look at it - and it looks amazing.

A lot of great thoughts and effort put into it.

I would like to use it as a reference case for our Thursday session.

Would that be ok with you ?

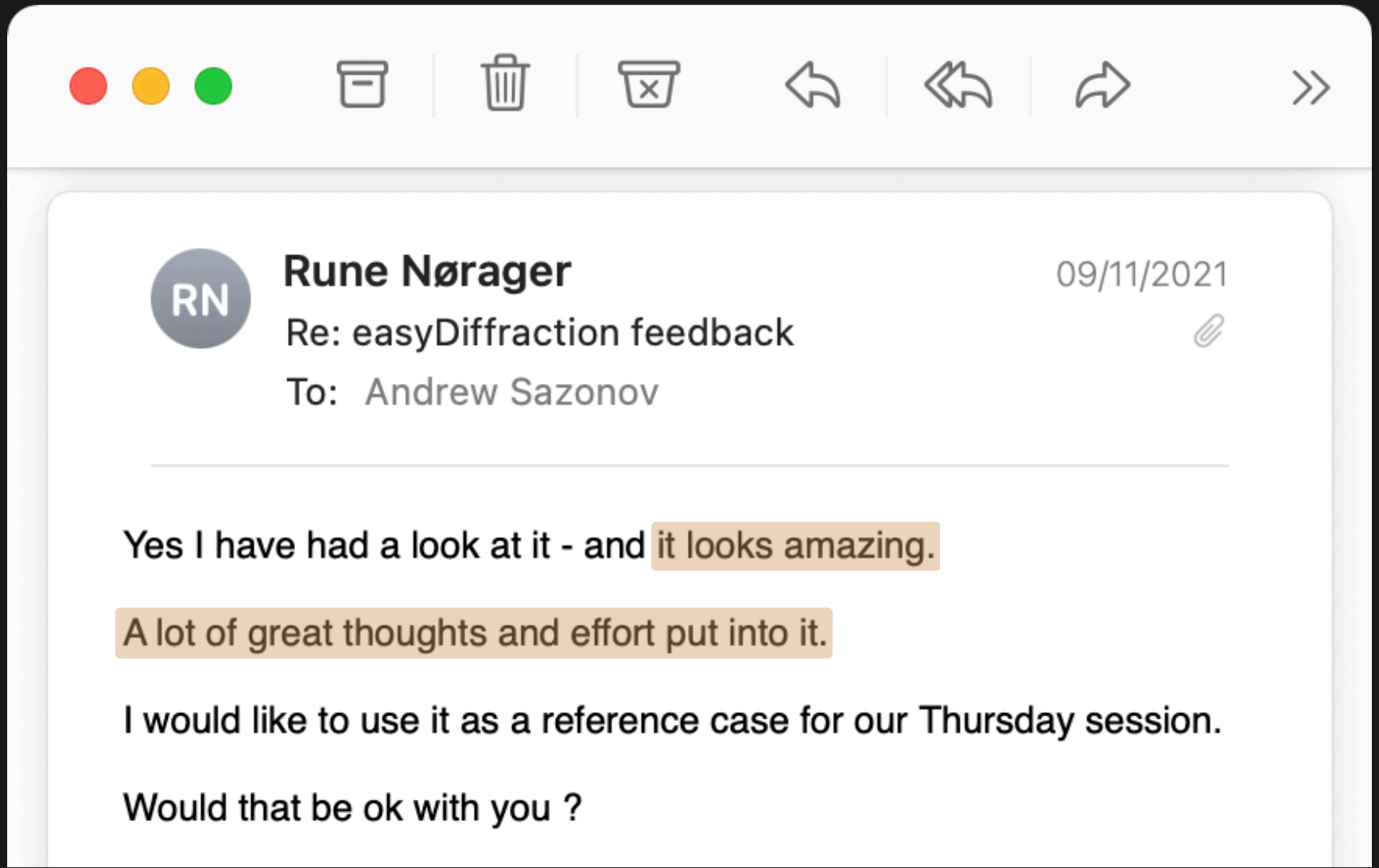
Kind regards.

Rune Nørager, PhD
Adfærds- & designpsykolog, CEO

DESIGN PSYCHOLOGY

Reviewed by UX experts

Then they reviewed our interface.



RN

Rune Nørager 09/11/2021

Re: easyDiffraction feedback

To: Andrew Sazonov

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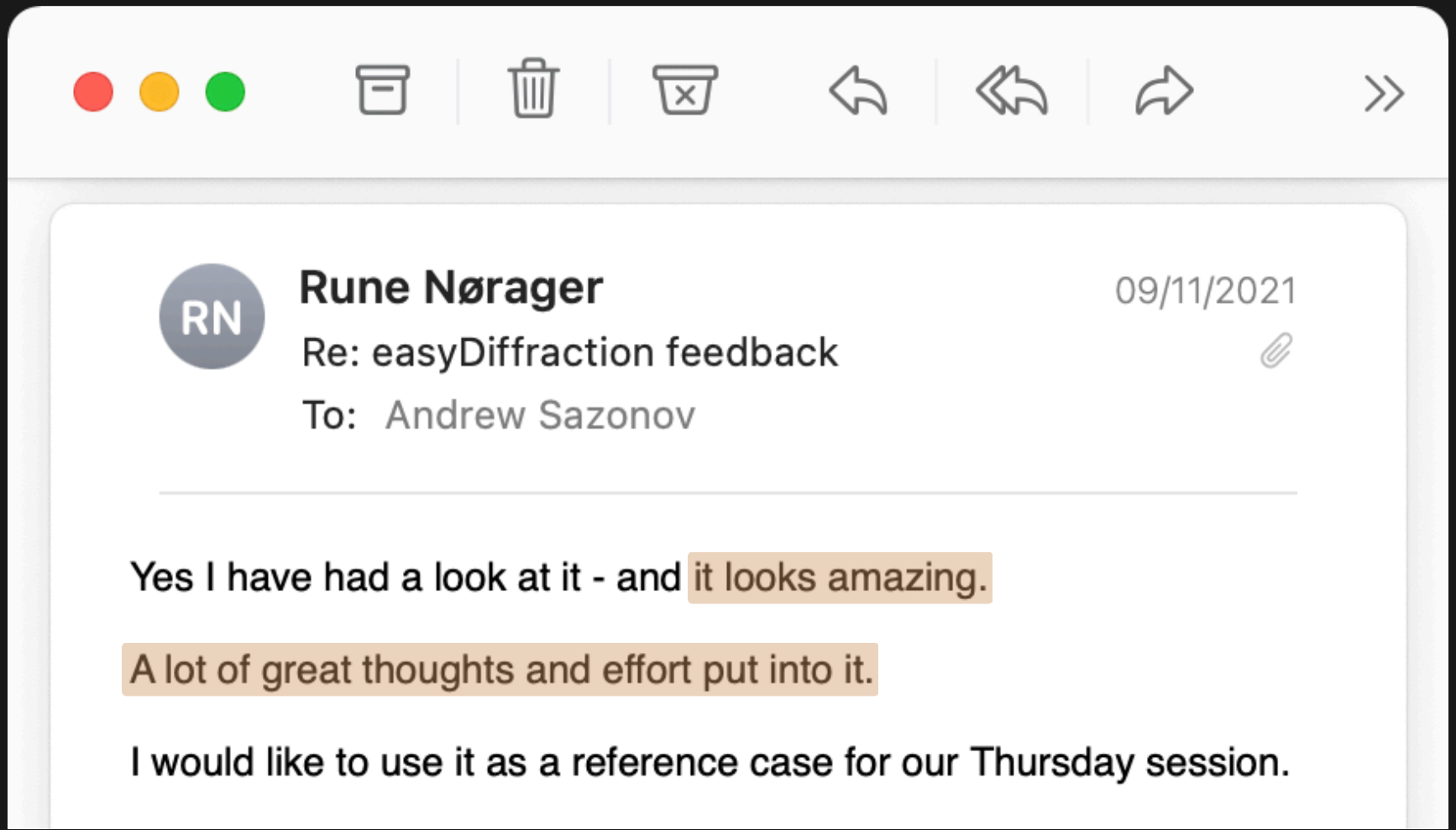
I would like to use it as a reference case for our Thursday session.

Would that be ok with you ?

Reviewed by UX experts

Their verdict: a well-designed interface.

Then they reviewed our interface.



RN

Rune Nørager 09/11/2021

Re: easyDiffraction feedback

To: Andrew Sazonov

Yes I have had a look at it - and it looks amazing.

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2.6 EASYDIFFRACTION

Three ways to run the same workflow

2.6 EASYDIFFRACTION

Three ways to run the same workflow

Graphical interface

for users who prefer to click



2.6 EASYDIFFRACTION

Three ways to run the same workflow

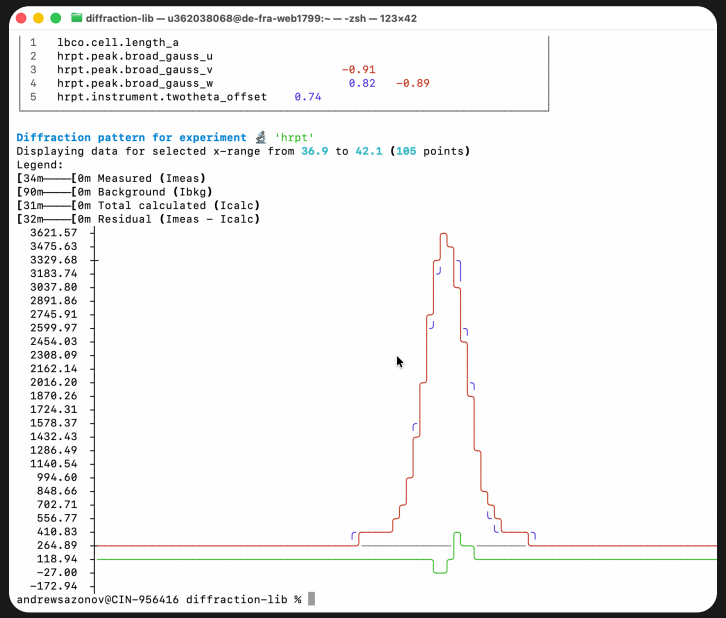
Graphical interface

for users who prefer to click



Command line

for command-line enthusiasts



2.6 EASYDIFFRACTION

Three ways to run the same workflow

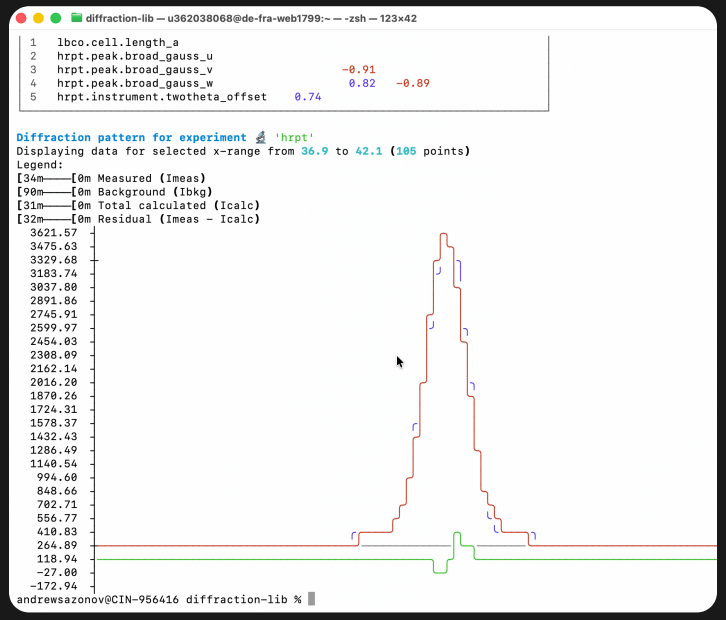
Graphical interface

for users who prefer to click



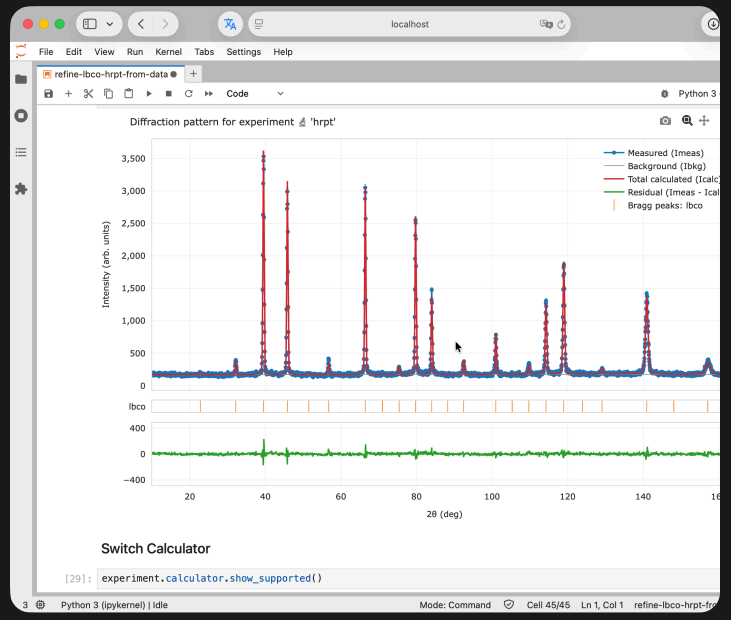
Command line

for command-line enthusiasts



Python library

for those who prefer scripts/Jupyter

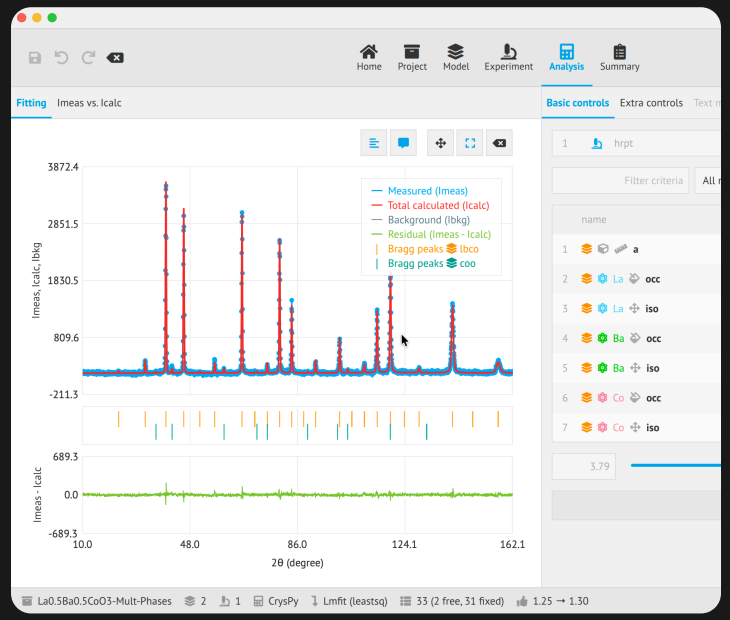


Three ways to run the same workflow

The same project, whichever way you open it.

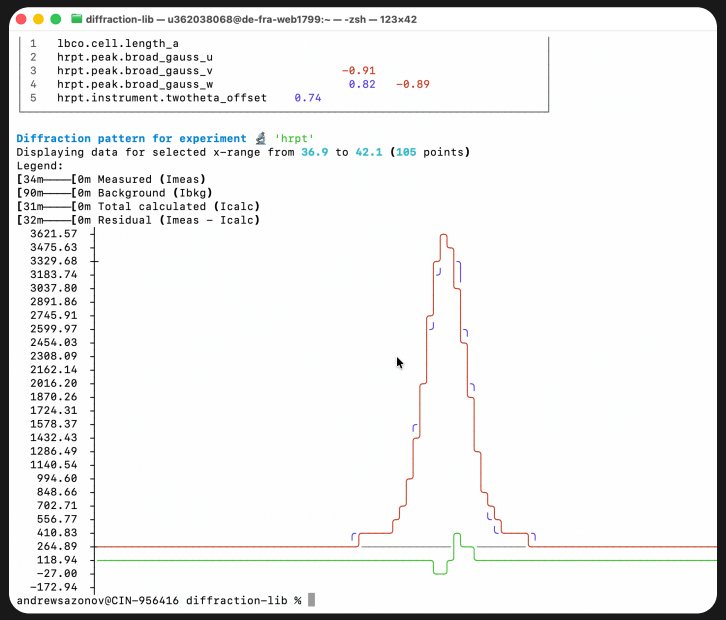
Graphical interface

for users who prefer to click



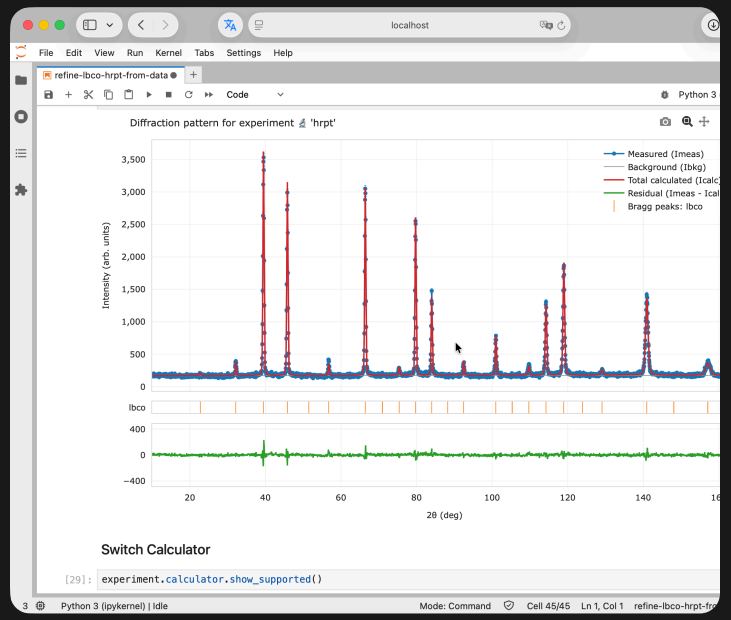
Command line

for command-line enthusiasts



Python library

for those who prefer scripts/Jupyter



2.7 EASYDIFFRACTION

Same look and feel, different technique

Same look and feel, different technique

EasyDiffraction

a powder pattern and a crystal structure



Same look and feel, different technique

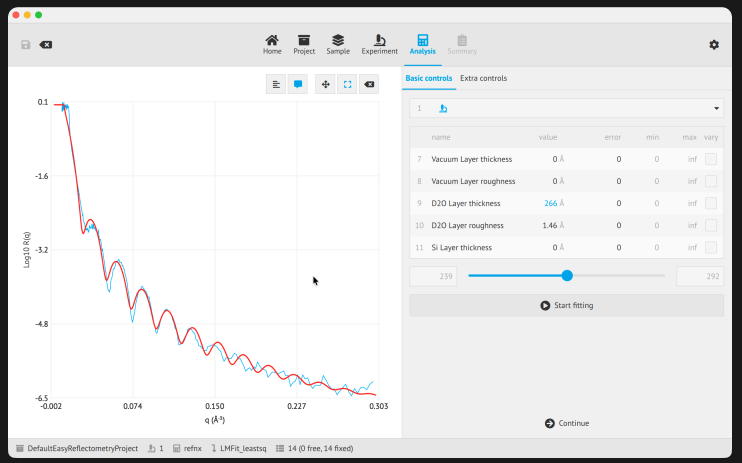
EasyDiffraction

a powder pattern and a crystal structure



EasyReflectometry

a reflectivity curve and a layer stack



Same look and feel, different technique

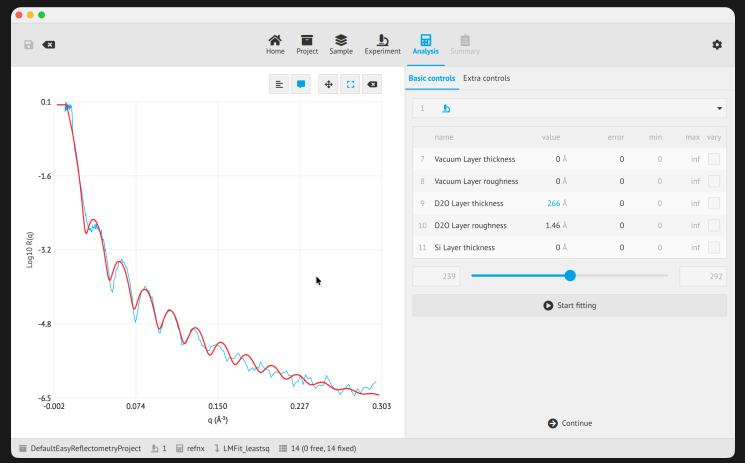
EasyDiffraction

a powder pattern and a crystal structure



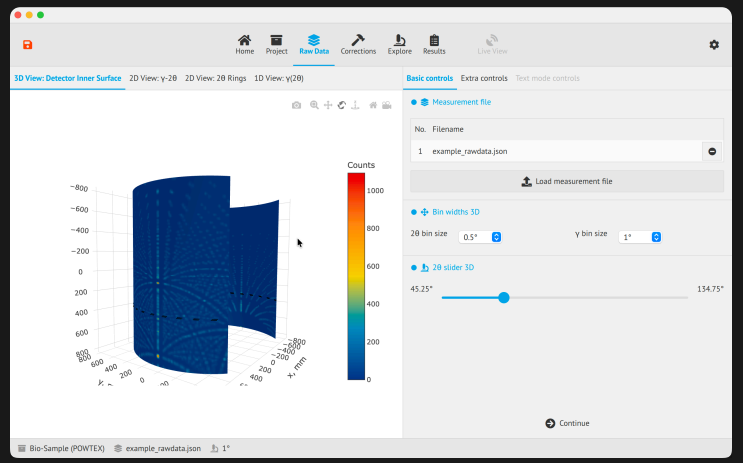
EasyReflectometry

a reflectivity curve and a layer stack



EasyTexture

raw detector data, before any model



Same look and feel, different technique

The same layout works for other techniques, even without fitting.

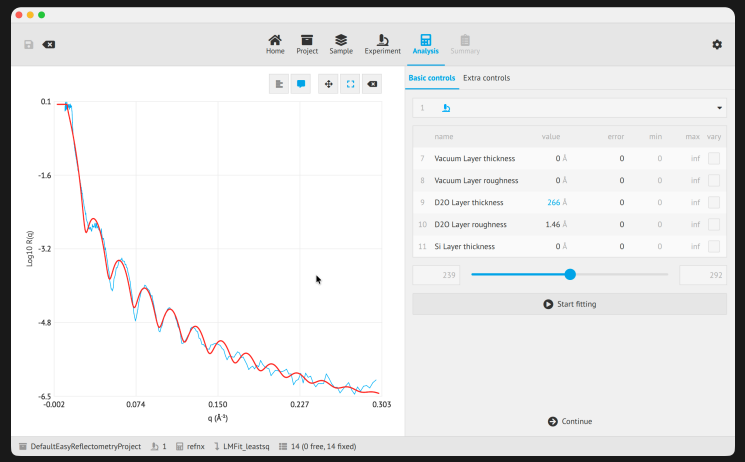
EasyDiffraction

a powder pattern and a crystal structure



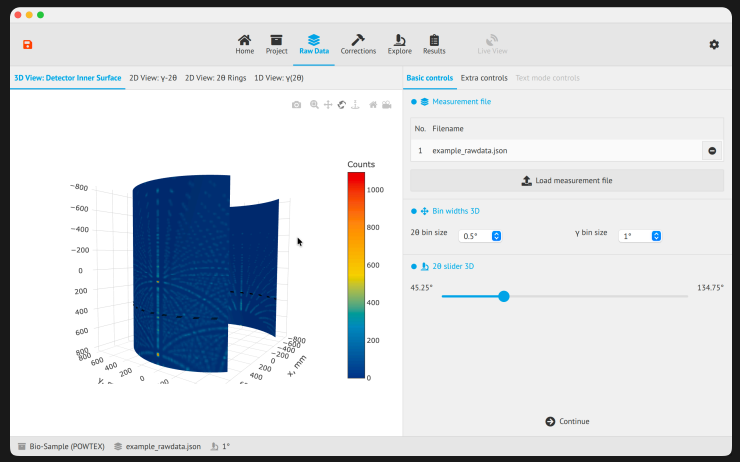
EasyReflectometry

a reflectivity curve and a layer stack



EasyTexture

raw detector data, before any model



2.8 EASYDIFFRACTION

What became reusable

What became reusable

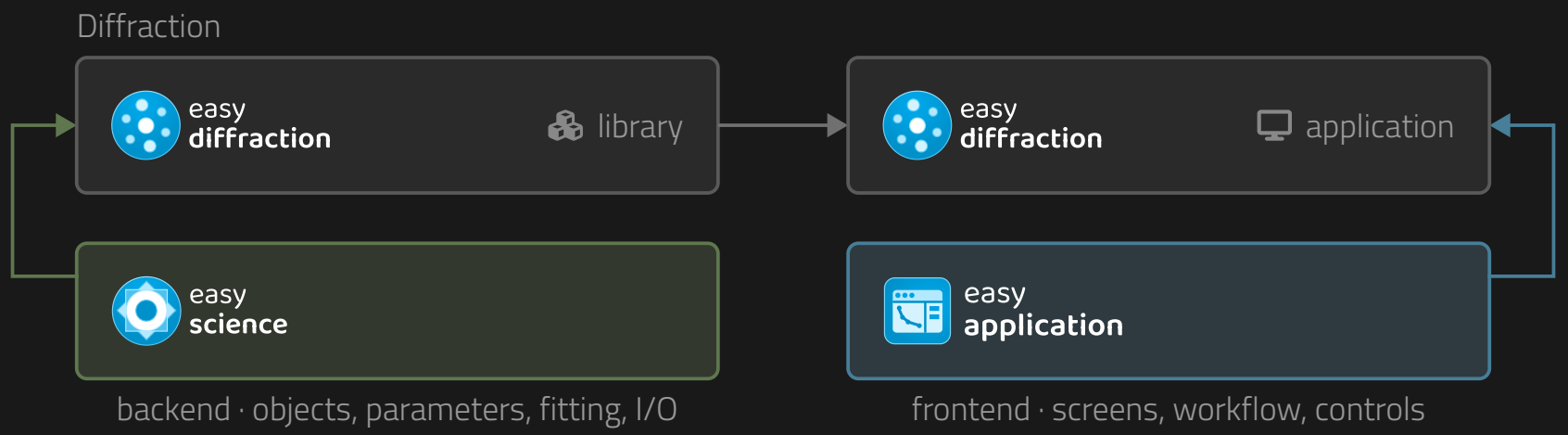


backend · objects, parameters, fitting, I/O

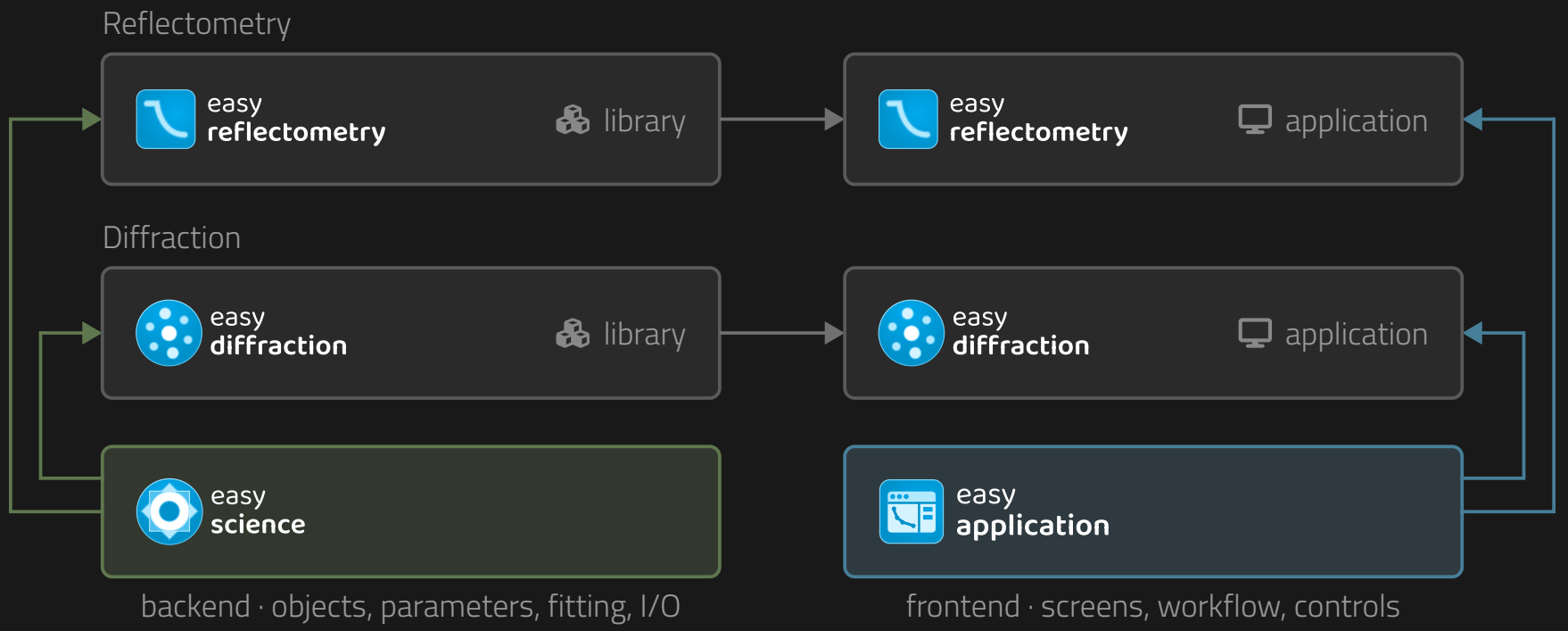


frontend · screens, workflow, controls

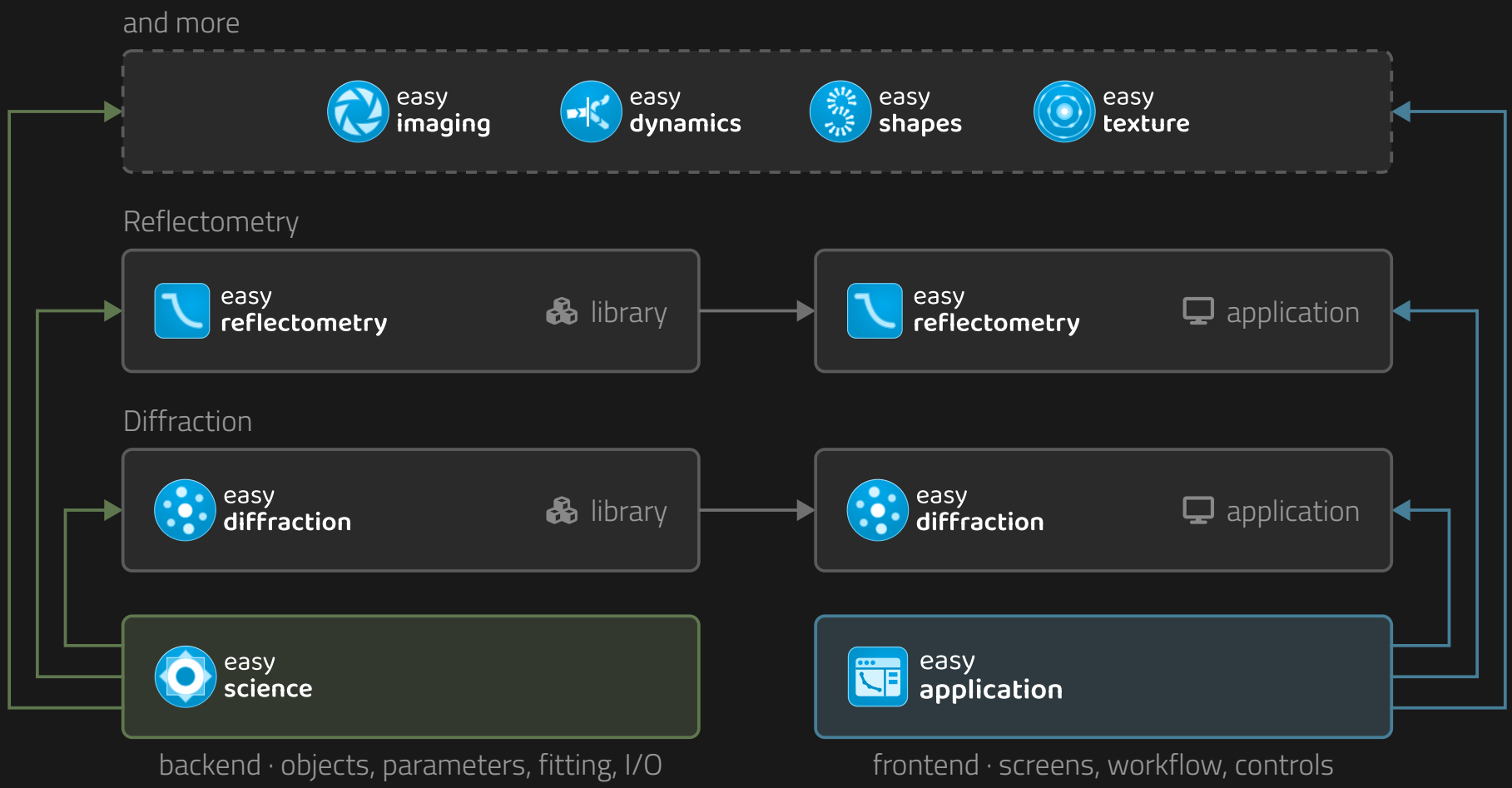
What became reusable



What became reusable

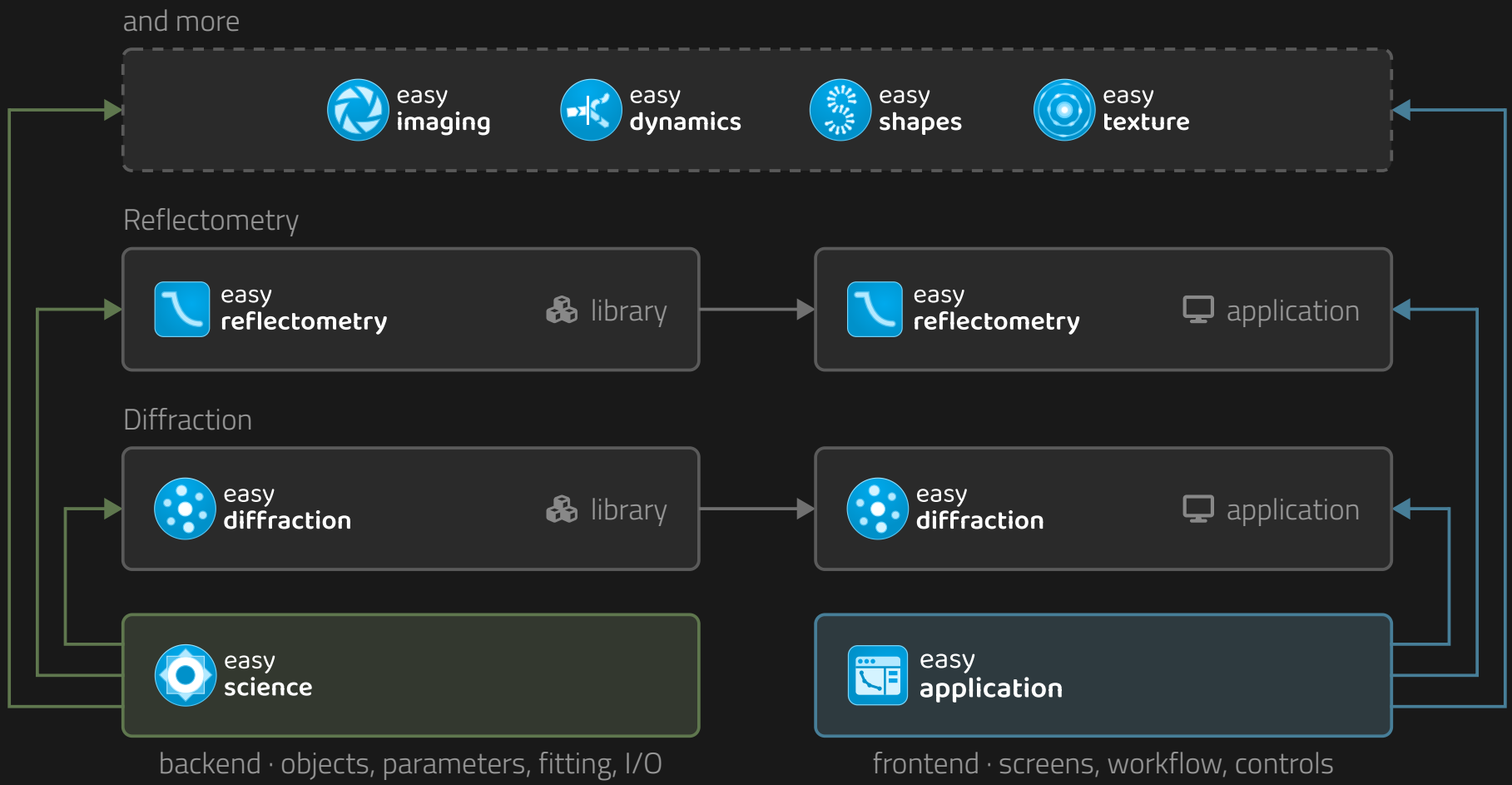


What became reusable



What became reusable

With EasyScience, a new technique adds only its science.



2.9 EASYDIFFRACTION

Where EasyScience is used today

Where EasyScience is used today

easyscience.org · github.com/easyscience



easy diffraction

analysis running

diffraction



easy reflectometry

analysis running

reflectometry

Where EasyScience is used today

easyscience.org · github.com/easyscience

easy diffraction — analysis — running

diffraction

easy reflectometry — analysis — running

reflectometry

easy dynamics — analysis — new

quasielastic neutron scattering

easy imaging — analysis — new

Bragg-edge imaging

2.9 EASYDIFFRACTION

Where EasyScience is used today

easyscience.org · github.com/easyscience



easy diffraction

analysis · running

diffraction



easy reflectometry

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reflectometry



easy dynamics

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easy imaging

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Bragg-edge imaging



easy texture

reduction · external

FZ Jülich · Göttingen

texture

Where EasyScience is used today

easyscience.org · github.com/easyscience

easy diffraction
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easy imaging
Bragg-edge imaging
analysis · new

easy texture
texture
reduction · external · FZ Jülich · Göttingen

easy shapes
molecular structures
modelling · new · Shapespyre · Chalmers

Where EasyScience is used today

Projects at different stages, some with partners.

easyscience.org · github.com/easyscience

easy diffraction
analysis · running
diffraction

easy reflectometry
analysis · running
reflectometry

easy dynamics
analysis · new
quasielastic neutron scattering

easy imaging
analysis · new
Bragg-edge imaging

easy texture
reduction · external
FZ Jülich · Göttingen
texture

easy shapes
modelling · new
Shapespyre · Chalmers
molecular structures

1 WHY NEW SOFTWARE

2 EASYDIFFRACTION

3 CRYSTA

4 STATUS AND PLANS

3.1 CRYSTA

Back to diffraction: what analysis needs

3.1 CRYSTA

Back to diffraction: what analysis needs

Large series · 1,000 datasets × 100 patterns · temperature, field, time

3.1 CRYSTA

Back to diffraction: what analysis needs

Large series · 1,000 datasets × 100 patterns · temperature, field, time

Multidimensional data · huge parameter spaces · 2D and 3D detectors

3.1 CRYSTA

Back to diffraction: what analysis needs

- **Large series** · 1,000 datasets × 100 patterns · temperature, field, time
- **Multidimensional data** · huge parameter spaces · 2D and 3D detectors
- **Automation** · experiments that run themselves · no human in the loop

3.1 CRYSTA

Back to diffraction: what analysis needs

EasyDiffraction works today, but not fast enough.

- Large series · 1,000 datasets × 100 patterns · temperature, field, time
- Multidimensional data · huge parameter spaces · 2D and 3D detectors
- Automation · experiments that run themselves · no human in the loop

3.2 CRYSTA

AI in crystallography

3.2 CRYSTA

AI in crystallography



27th Congress and General Assembly
of the International Union of Crystallography

Calgary, August 2026 · ≈ 1,800 attendees

3.2 CRYSTA

AI in crystallography



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AI or machine learning named in the title or abstract

3.2 CRYSTA

AI in crystallography



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AI or machine learning named in the title or abstract

23
of 142 microsymposia

3.2 CRYSTA

AI in crystallography



27th Congress and General Assembly
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AI or machine learning named in the title or abstract

23
of 142 microsymposia

121
of 839 microsymposium talks

3.2 CRYSTA

AI in crystallography



27th Congress and General Assembly
of the International Union of Crystallography

Calgary, August 2026 · ≈ 1,800 attendees

AI or machine learning named in the title or abstract

23	121	69
of 142 microsymposia	of 839 microsymposium talks	of 932 posters

AI in crystallography

AI is now widely used, from coding to experiment orchestration.



27th Congress and General Assembly
of the International Union of Crystallography

Calgary, August 2026 · ≈ 1,800 attendees

AI or machine learning named in the title or abstract

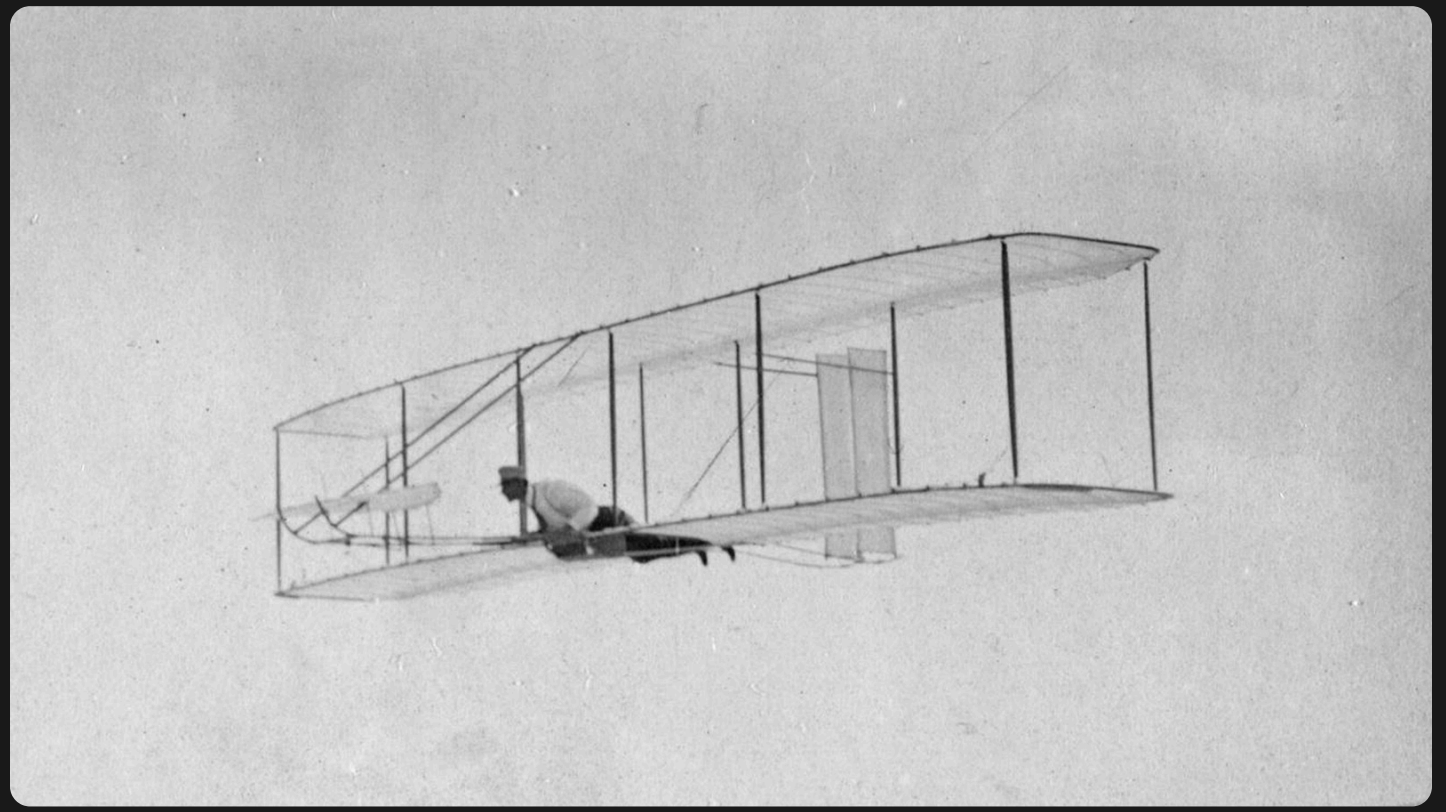
23	121	69
of 142 microsymposia	of 839 microsymposium talks	of 932 posters

3.3 CRYSTA

A pilot project

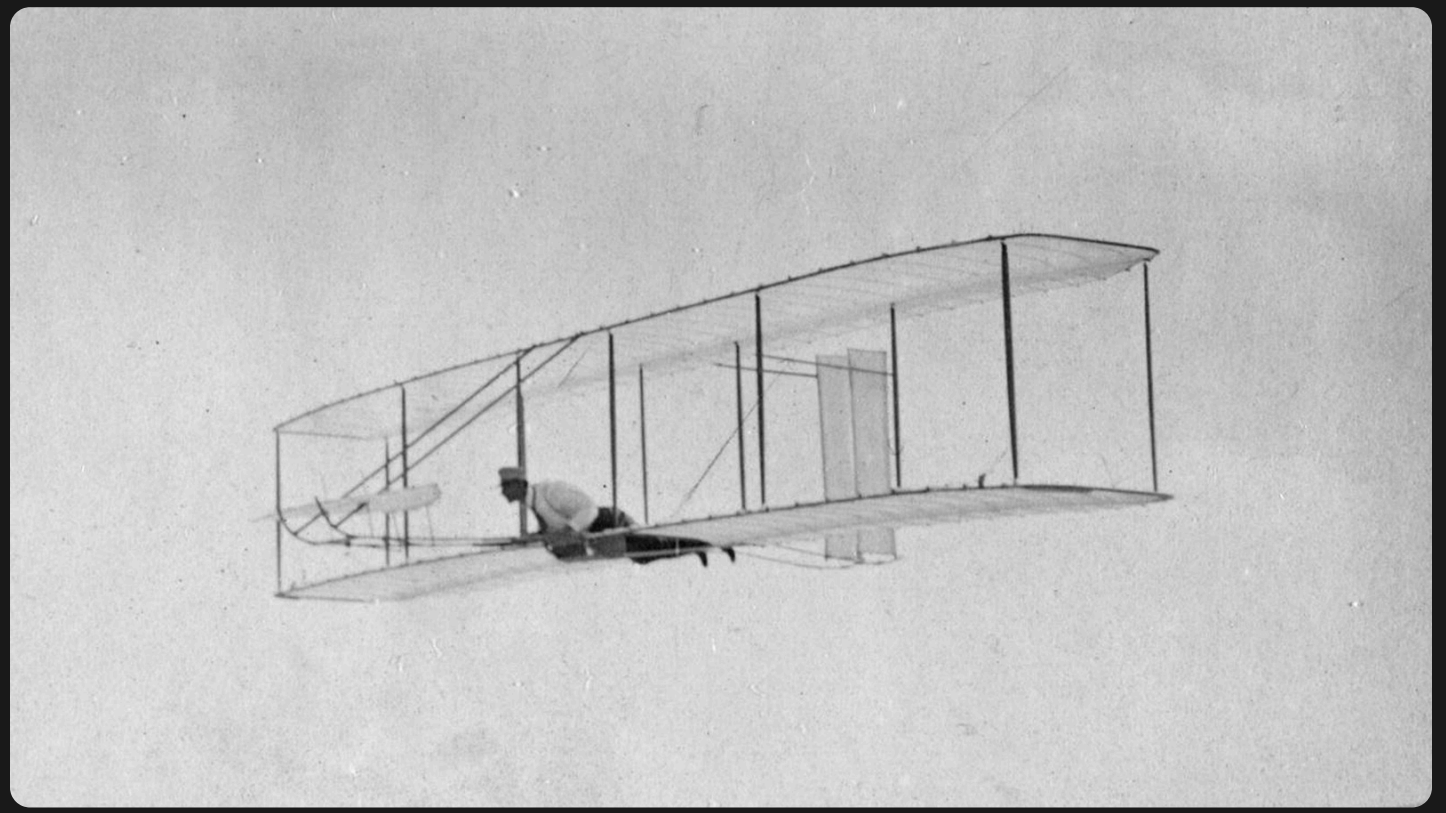
3.3 CRYSTA

A pilot project



A pilot project

Can supervised AI build a fast crystallographic library?



3.4 CRYSTA

Meet crysta

3.4 CRYSTA

Meet crysta

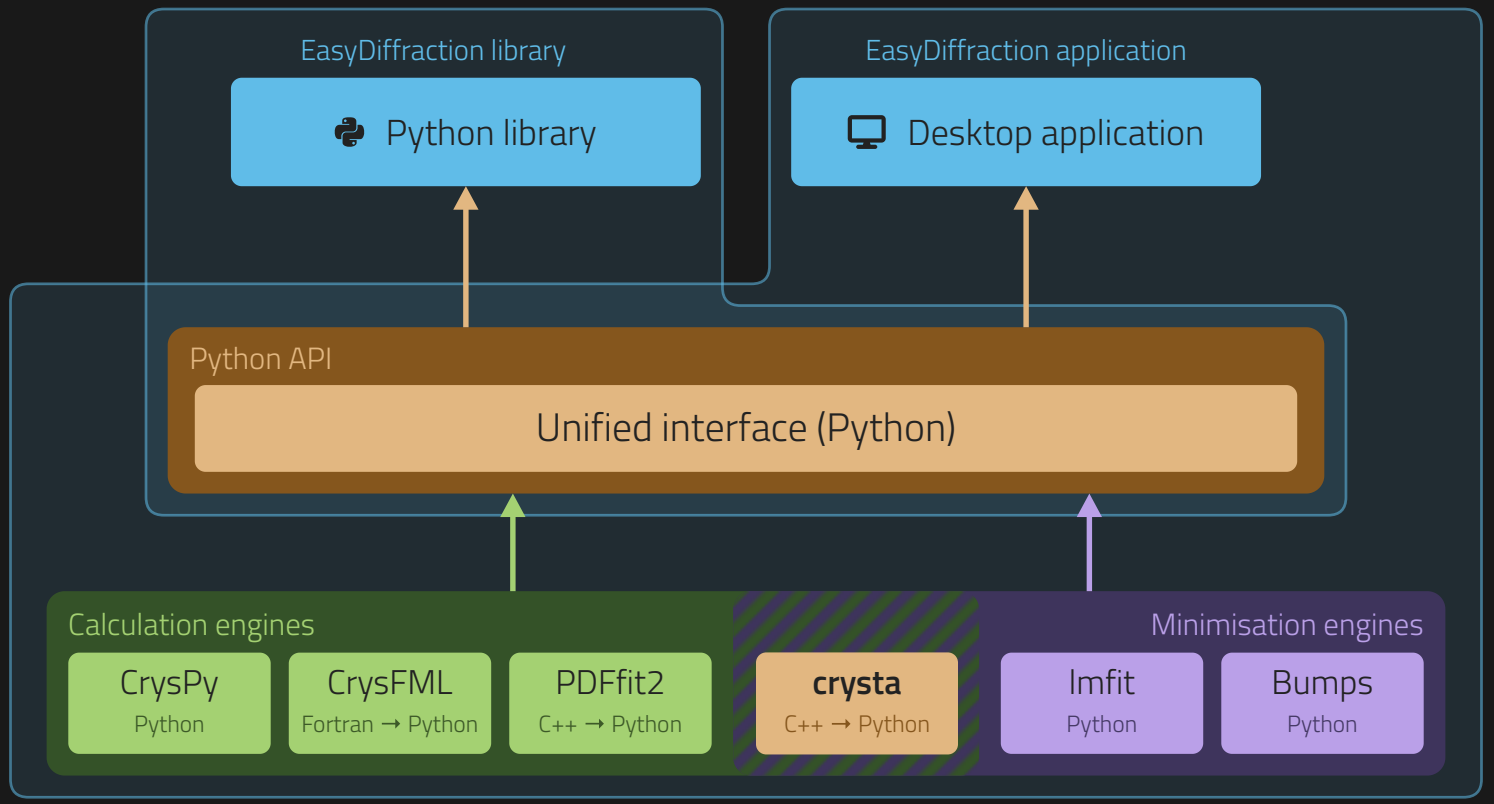
crysta

Eigen · nanobind · CMake



Meet crysta

A new calculation and minimisation engine, built in-house.



3.5 CRYSTA

The test case

The test case

Sample · NCAF, $Na_2Ca_3Al_2F_{14}$ · reference powder, cubic

The test case

Sample · NCAF, $Na_2Ca_3Al_2F_{14}$ · reference powder, cubic

Measurement · WISH at ISIS, banks 2 + 9 · 3977 points, 627 peaks

The test case

Sample · NCAF, $Na_2Ca_3Al_2F_{14}$ · reference powder, cubic

Measurement · WISH at ISIS, banks 2 + 9 · 3977 points, 627 peaks

Refinement · 48 free parameters · structure, profile, background

The test case

- Sample · NCAF, $Na_2Ca_3Al_2F_{14}$ · reference powder, cubic
- Measurement · WISH at ISIS, banks 2 + 9 · 3977 points, 627 peaks
- Refinement · 48 free parameters · structure, profile, background
- Method · weighted least squares · same start, $\chi^2 \approx 4.8$ for all

The test case

- Sample · NCAF, $Na_2Ca_3Al_2F_{14}$ · reference powder, cubic
- Measurement · WISH at ISIS, banks 2 + 9 · 3977 points, 627 peaks
- Refinement · 48 free parameters · structure, profile, background
- Method · weighted least squares · same start, $\chi^2 \approx 4.8$ for all
- Reference · FullProf 8.40 · same model, one thread

The test case

One real case, the same for every code.

- **Sample** · NCAF, $Na_2Ca_3Al_2F_{14}$ · reference powder, cubic
- **Measurement** · WISH at ISIS, banks 2 + 9 · 3977 points, 627 peaks
- **Refinement** · 48 free parameters · structure, profile, background
- **Method** · weighted least squares · same start, $\chi^2 \approx 4.8$ for all
- **Reference** · FullProf 8.40 · same model, one thread

3.6 CRYSTA

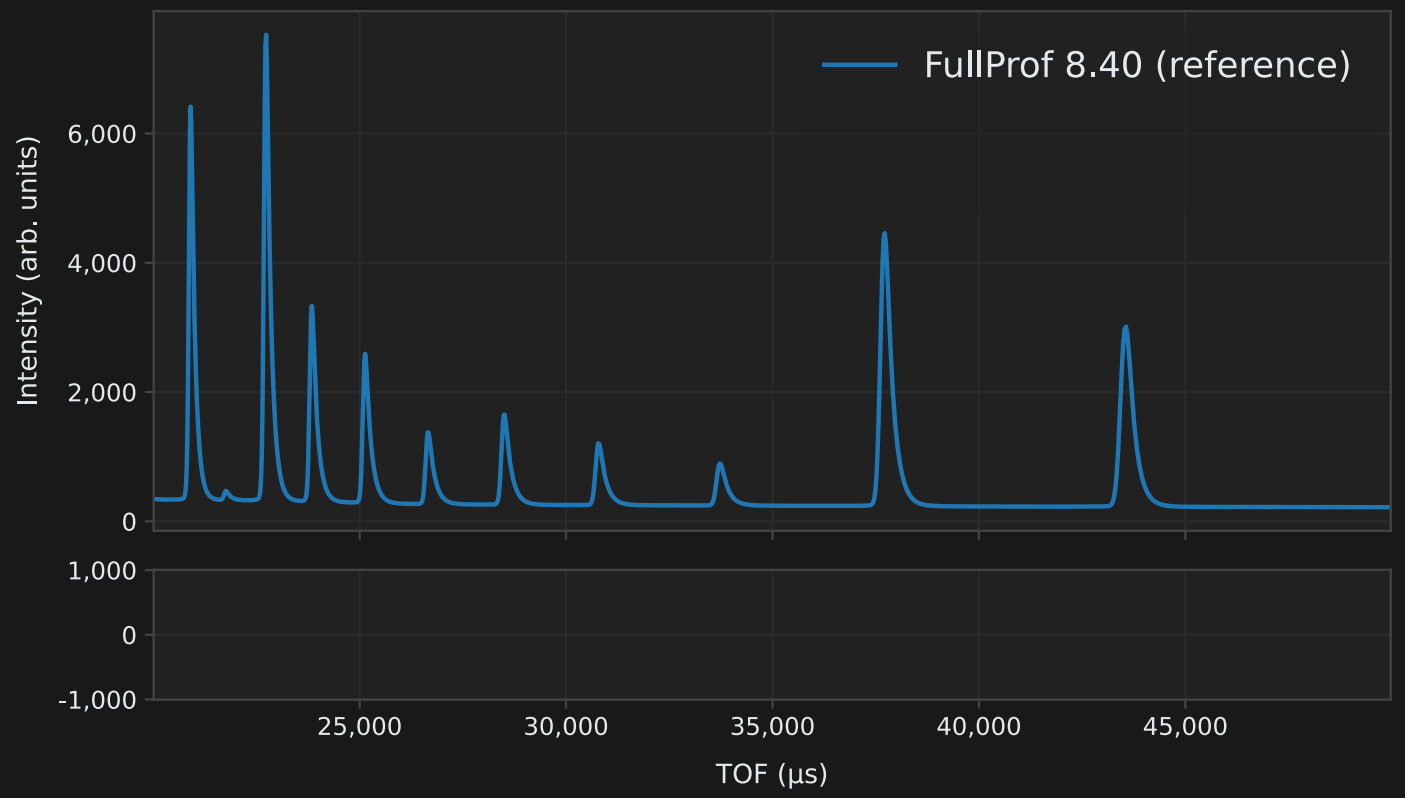
Agreement of calculated patterns

Agreement of calculated patterns

NCAF | WISH 2+9 | 3977 points | 627 peaks

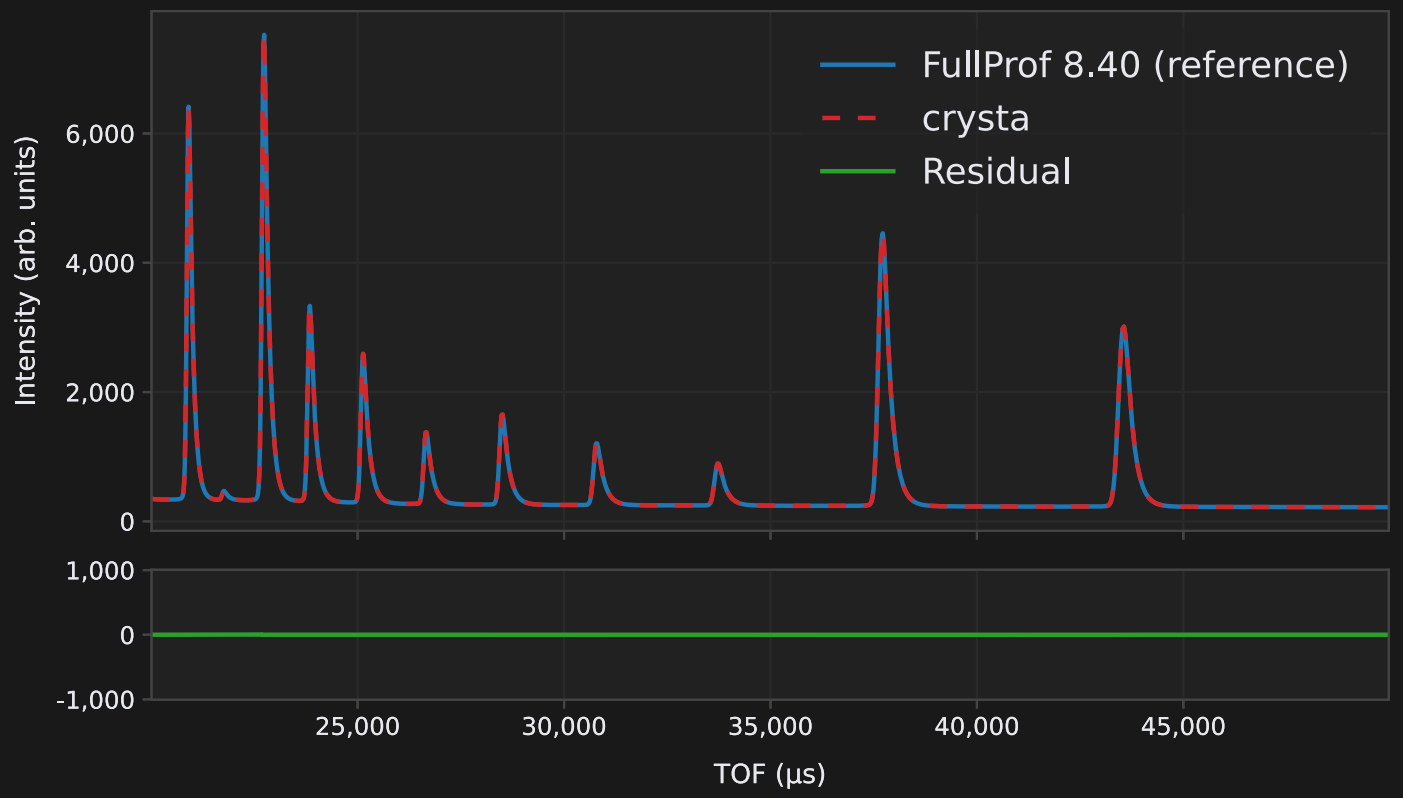
Agreement of calculated patterns

NCAF | WISH 2+9 | 3977 points | 627 peaks



Agreement of calculated patterns

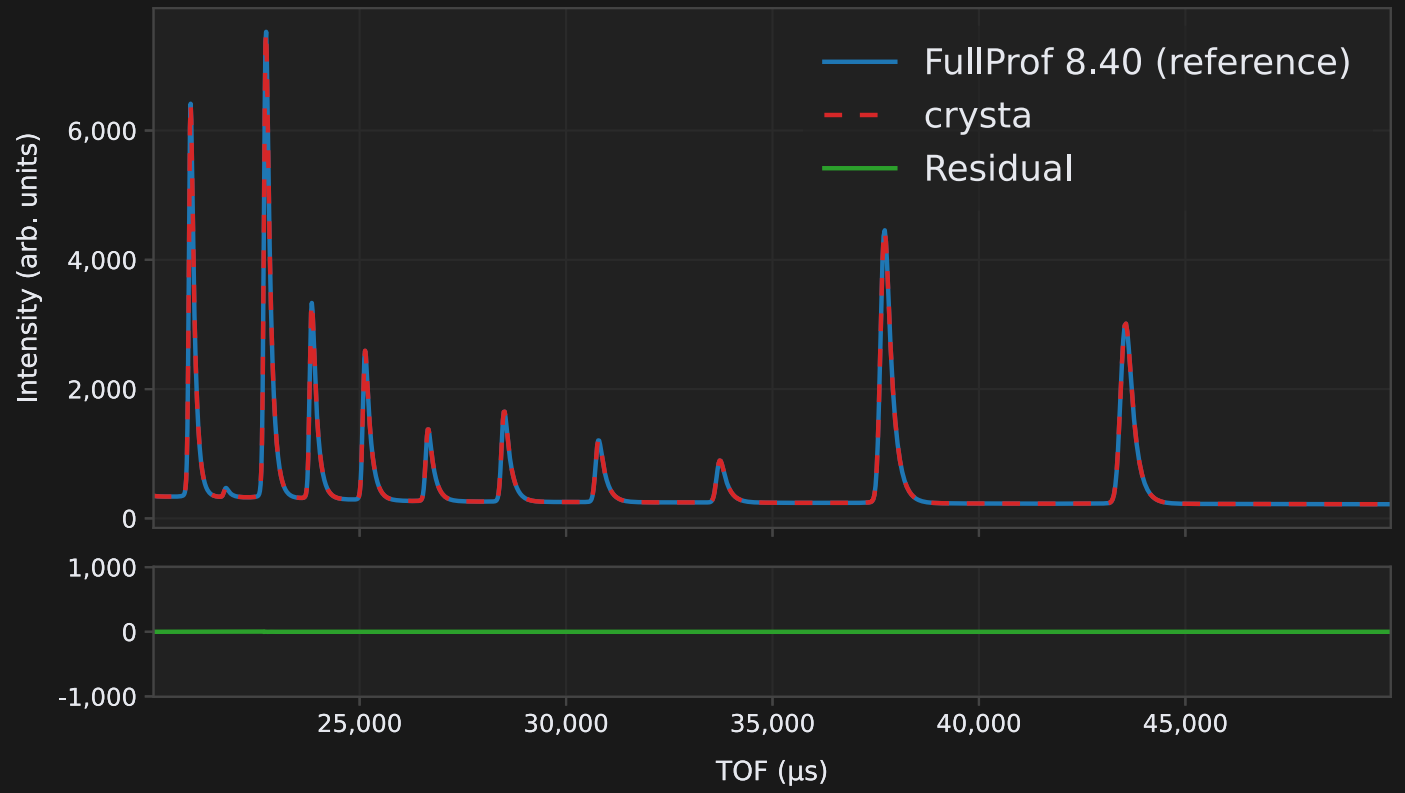
NCAF | WISH 2+9 | 3977 points | 627 peaks



Agreement of calculated patterns

Largest difference 3.9 of 7527 counts: 0.05 %.

NCAF | WISH 2+9 | 3977 points | 627 peaks



3.7 CRYSTA

Agreement of refined parameters

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Agreement of refined parameters

NCAF | WISH 2+9 | 3977 points | 627 peaks | same start | $\chi^2 \approx 4.8$ | 48 parameters

3.7 CRYSTA

Agreement of refined parameters

NCAF · WISH 2+9 · 3977 points · 627 peaks · same start · $\chi^2 \approx 4.8$ · 48 parameters

parameter	crysta	FullProf	Δ	Δ/σ
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Ca1 B_{iso}	1.12 ± 0.13	1.12 ± 0.13	−0.0017	0.01
F3 B_{iso}	1.85 ± 0.16	1.85 ± 0.17	−0.0004	0.00

Agreement of refined parameters

Every parameter within 0.69 σ of FullProf, median 0.05 σ.

NCAF | WISH 2+9 | 3977 points | 627 peaks | same start | $\chi^2 \approx 4.8$ | 48 parameters

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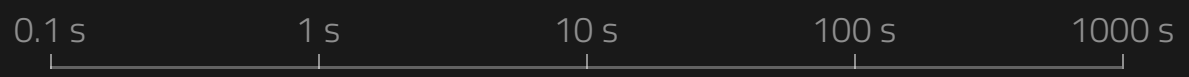
3.8 CRYSTA

Speed: a full refinement

3.8 CRYSTA

Speed: a full refinement

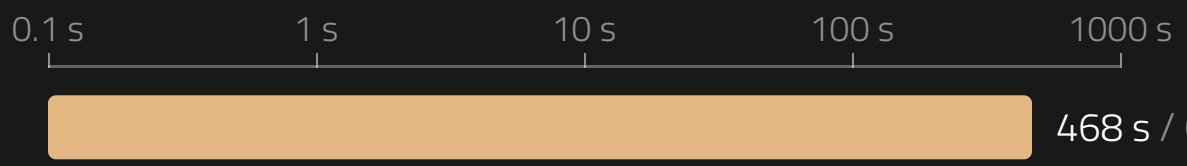
NCAF | WISH 2+9 | 3977 points | 627 peaks | 1 thread | log scale



3.8 CRYSTA

Speed: a full refinement

NCAF | WISH 2+9 | 3977 points | 627 peaks | 1 thread | log scale

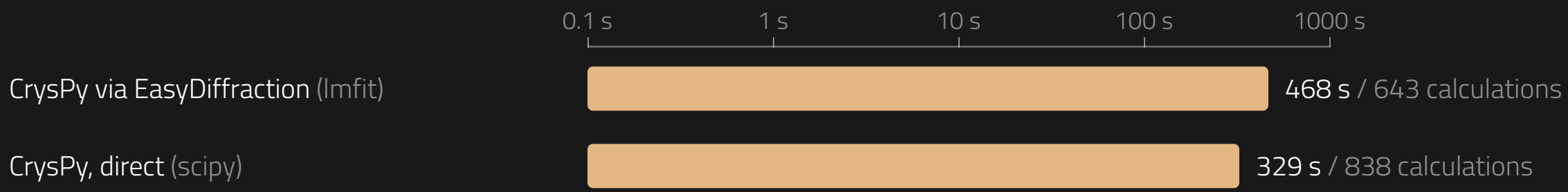


CrysPy via EasyDiffraction (lmfit) — 468 s / 643 calculations

3.8 CRYSTA

Speed: a full refinement

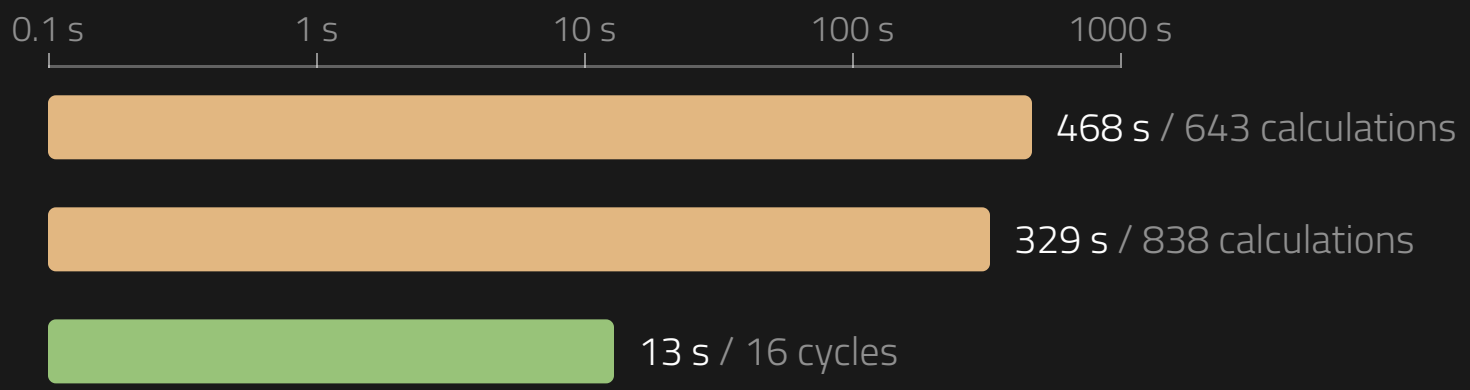
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3.8 CRYSTA

Speed: a full refinement

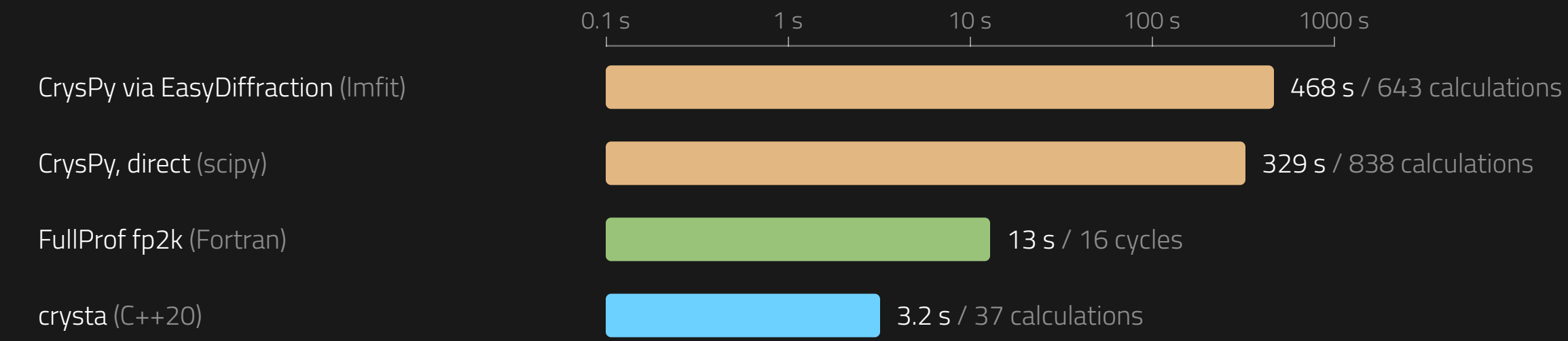
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3.8 CRYSTA

Speed: a full refinement

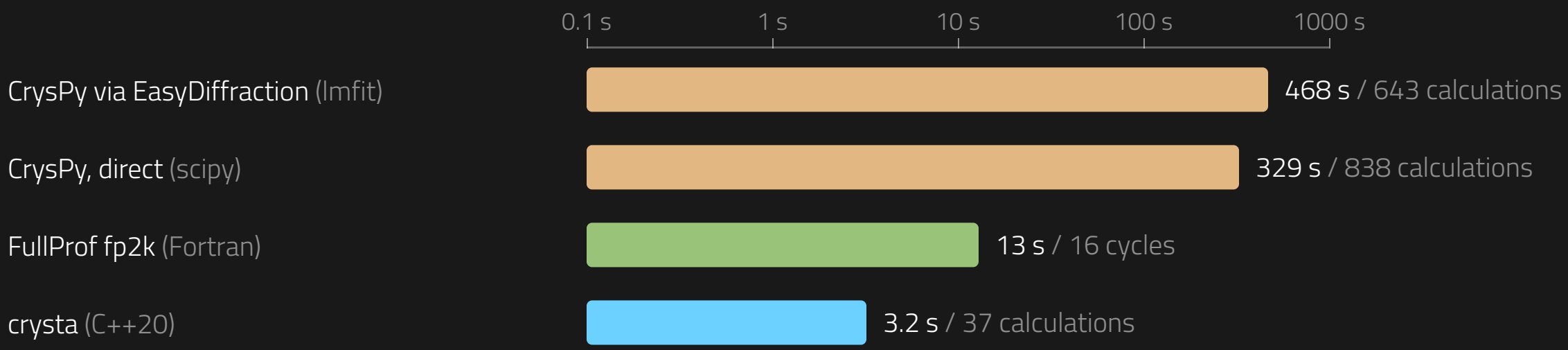
NCAF | WISH 2+9 | 3977 points | 627 peaks | 1 thread | log scale



Speed: a full refinement

150× faster than today's EasyDiffraction route.

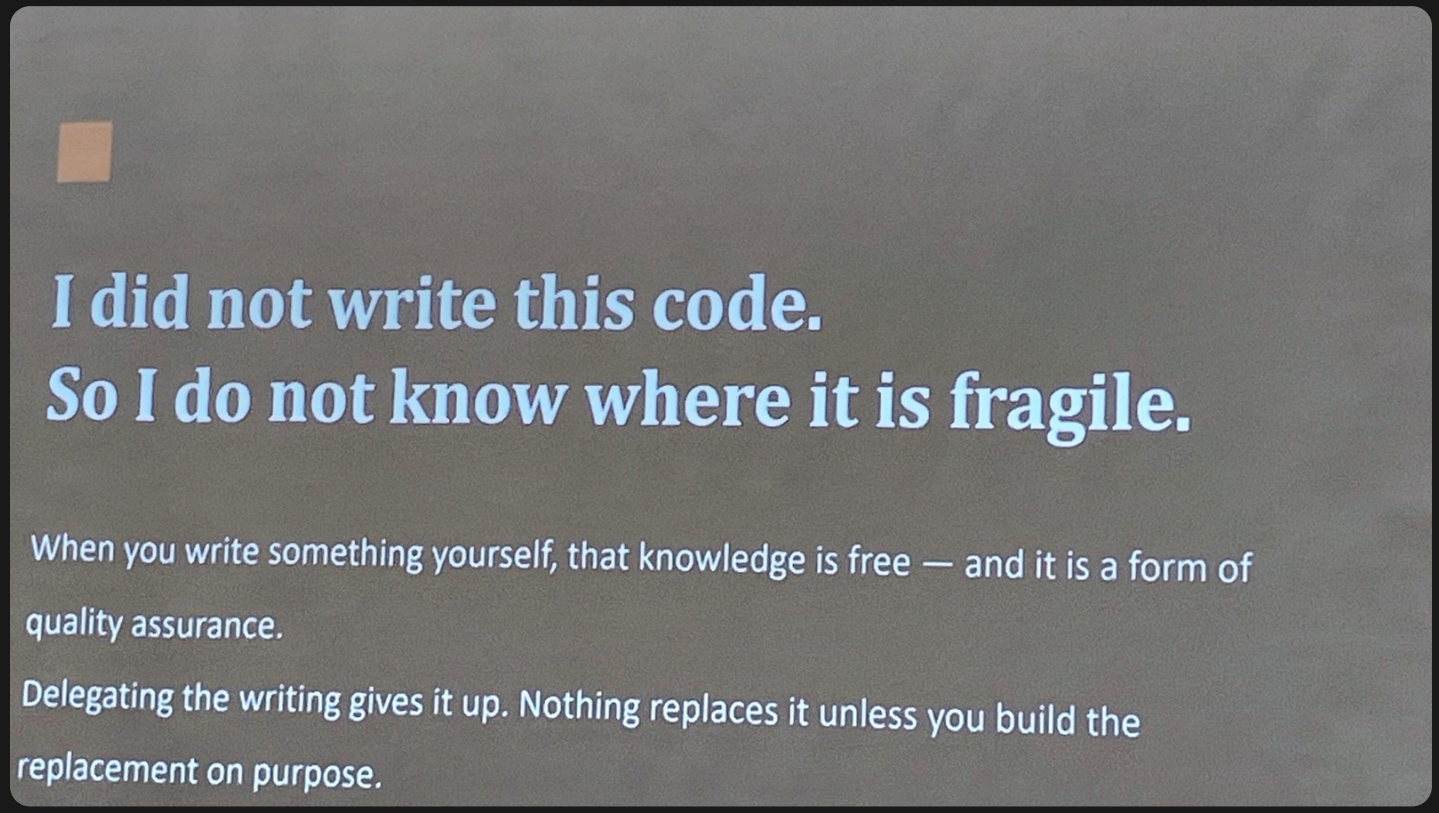
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3.9 CRYSTA

A lesson from IUCr 2026

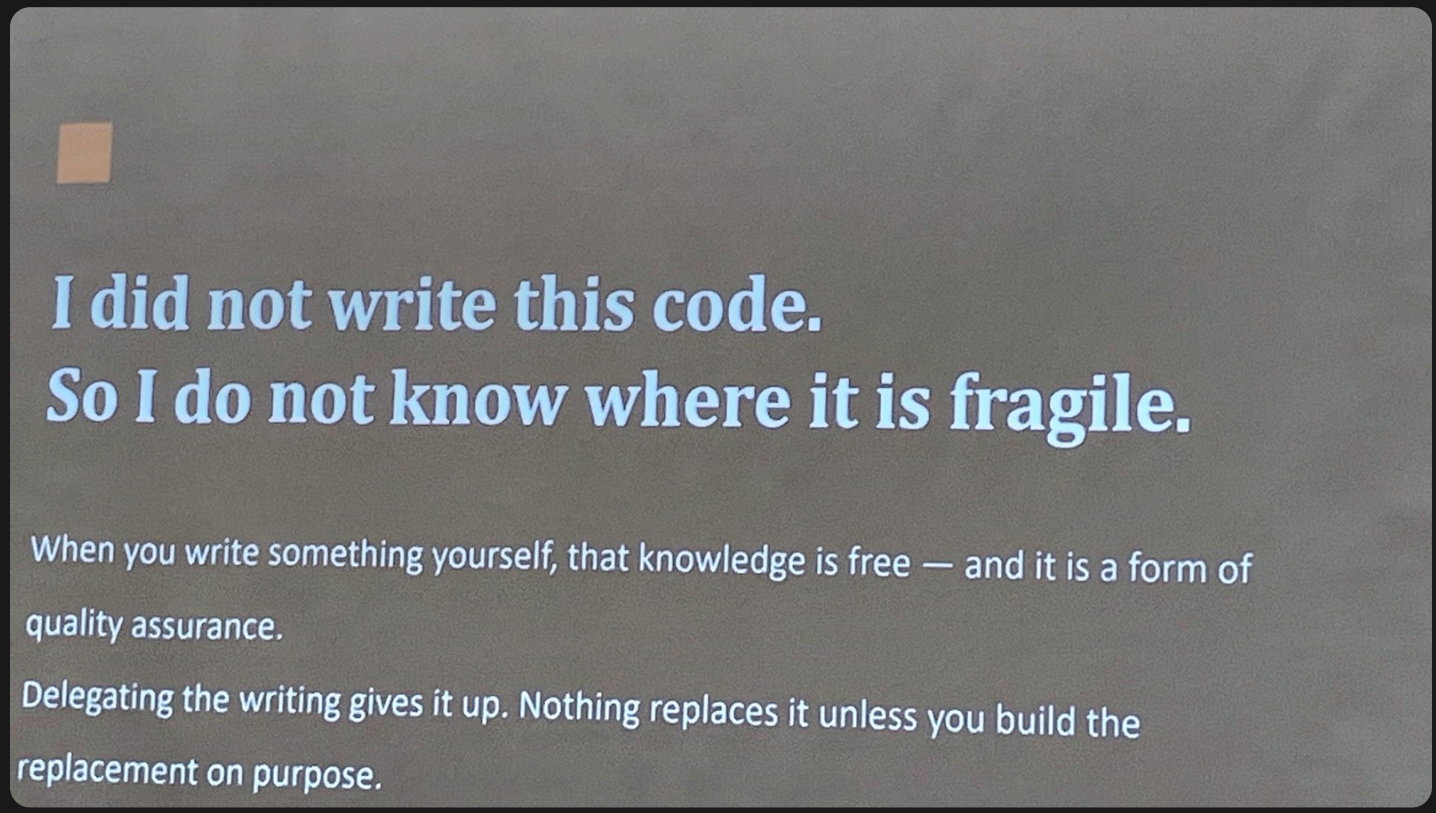
A lesson from IUCr 2026



Oleg Sobolev & Paul Adams, LBNL · AI-written CIF parser for cctbx · MS-116, IUCr 2026, Calgary

A lesson from IUCr 2026

Nobody knows where AI-written code is fragile until it is tested.



Oleg Sobolev & Paul Adams, LBNL · AI-written CIF parser for cctbx · MS-116, IUCr 2026, Calgary

3.10 CRYSTA

How the code is checked

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Tests at every level · unit, integration, system

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Independent references · FullProf, CrysFML, CrysPy

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Strict builds · warnings are errors, static analysis

How the code is checked

- **Tests** at every level · unit, integration, system
- **Independent references** · FullProf, CrysFML, CrysPy
- **Strict builds** · warnings are errors, static analysis
- **Continuous integration** · every change, Linux, macOS and Windows

How the code is checked

AI writes the code, so it is checked more, not less.

- **Tests** at every level · unit, integration, system
- **Independent references** · FullProf, CrysFML, CrysPy
- **Strict builds** · warnings are errors, static analysis
- **Continuous integration** · every change, Linux, macOS and Windows

3.11 CRYSTA

Credits

3.11 CRYSTA

Credits

LITERATURE

International Tables A and C · Cromer & Mann · Waasmaier & Kirfel · Busing & Levy · Caglioti · Thompson, Cox & Hastings · Finger, Cox & Jephcoat · Von Dreele · Ikeda & Carpenter · Becker & Coppens · Dollase · Rietveld · Marquardt · Proffen & Billinge · Egami & Billinge · Blume · Bertaut · Wills · Sears · Lovesey · Shmueli · Prince · Larson & Von Dreele · …

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OPEN-SOURCE SOFTWARE

CrysPy: Iurii Kibalin (ESS), Arsen Gukasov (ILL), Andrew Sazonov (ESS) · **CrysFML**: Juan Rodríguez-Carvajal (ILL), Javier González-Platas (ULL), Nebil Ayape Katcho (ILL) · **cctbx**: R. Grosse-Kunstleve, P. Adams, N. Sauter et al. (LBNL) · **GSAS-II**: B. Toby, R. Von Dreele (ANL) · **PDFfit2**: C. Farrow, P. Juhás, T. Proffen, S. Billinge et al. · …

3.11 CRYSTA

Credits

Decades of work by many people, most of it long before AI.

LITERATURE

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1 WHY NEW SOFTWARE

2 EASYDIFFRACTION

3 CRYSTA

4 STATUS AND PLANS

4.1 🏁 STATUS AND PLANS

How much is done

How much is done

EasyDiffraction



How much is done



How much is done

crysta after eight weeks: how far can it go?



4.2 🏁 STATUS AND PLANS

Next steps

4.2 🏁 STATUS AND PLANS

Next steps

- Magnetic structures · from CrysPy and CrysFML, the biggest gap

4.2 🏁 STATUS AND PLANS

Next steps

- **Magnetic structures** · from CrysPy and CrysFML, the biggest gap
- **Polarised neutrons** · from CrysPy

4.2 STATUS AND PLANS

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- **Magnetic structures** · from CrysPy and CrysFML, the biggest gap
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- **Library features in the application** · the library is two years ahead

4.2 STATUS AND PLANS

Next steps

- **Magnetic structures** · from CrysPy and CrysFML, the biggest gap
- **Polarised neutrons** · from CrysPy
- **Library features in the application** · the library is two years ahead
- **Fixes and extensions** · bugs, usability, small features

Next steps

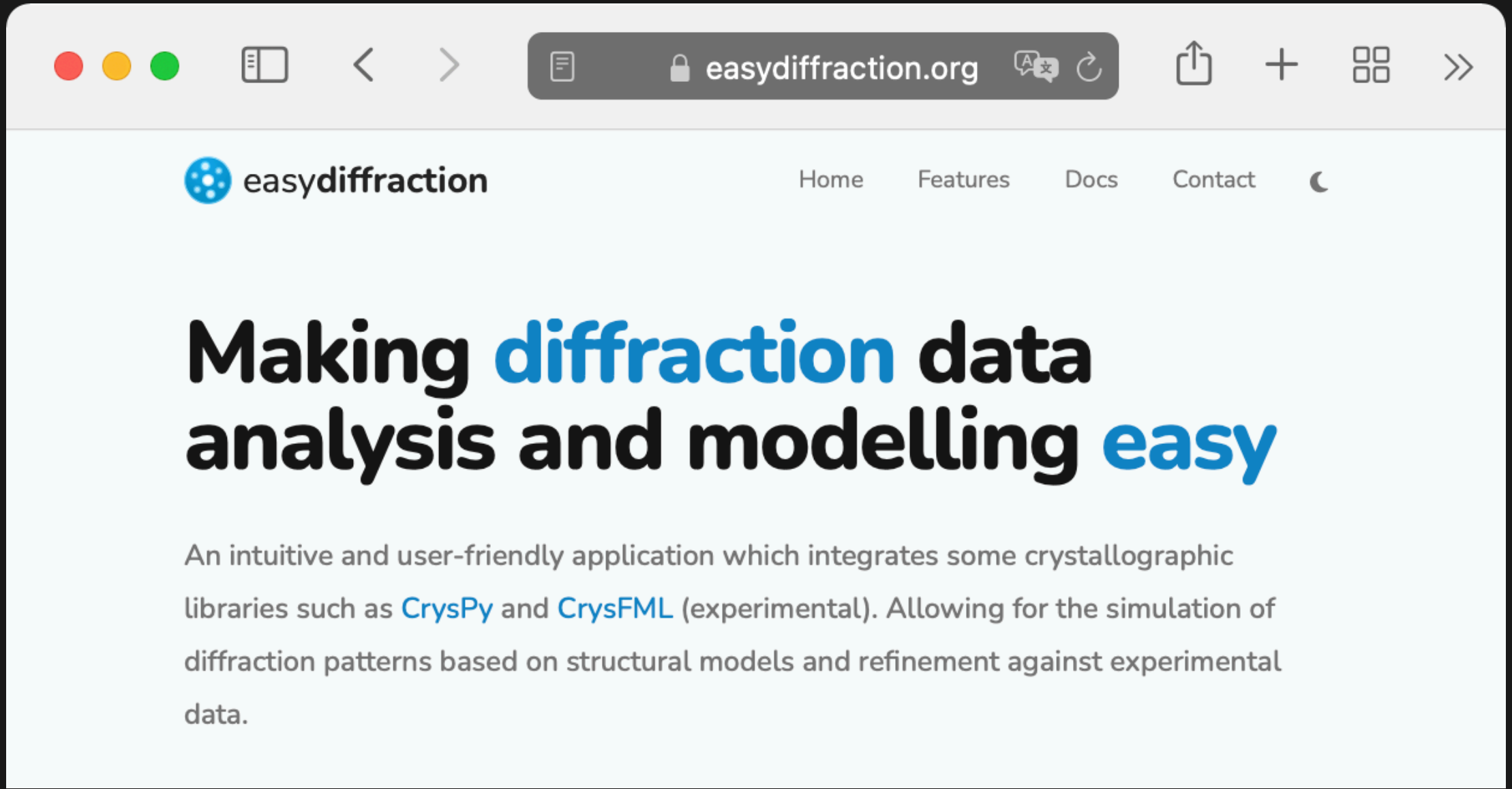
Bringing existing features to users.

- **Magnetic structures** · from CrysPy and CrysFML, the biggest gap
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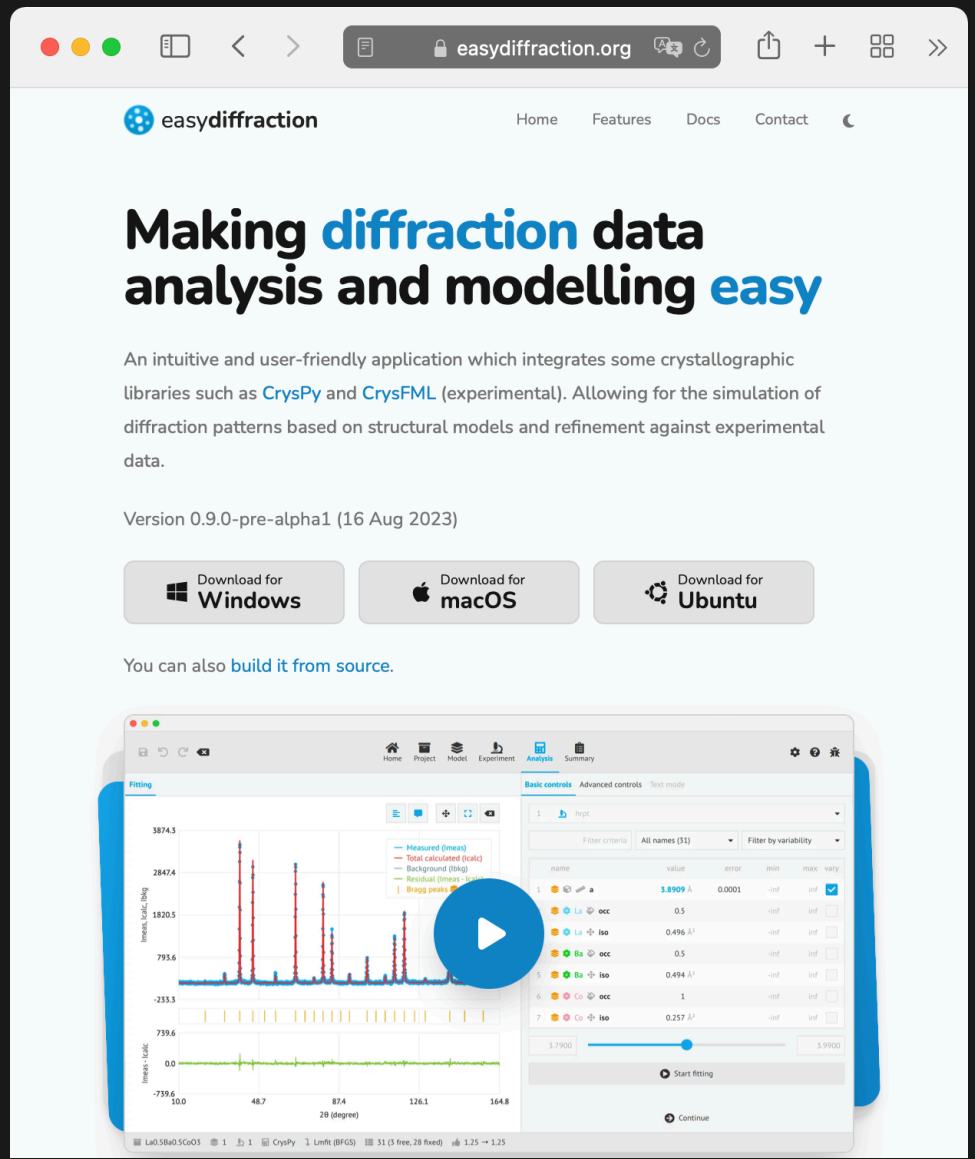
4.3 🏁 STATUS AND PLANS

Project page · easydiffraction.org

Project page · easydiffraction.org



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Documentation

Library

The EasyDiffraction library documentation covers everything related to using EasyDiffraction as a python library.

Application

The EasyDiffraction application documentation is the place to see how to use EasyDiffraction graphical user interface.

Get in touch

Project page · easydiffraction.org

